

smote

November 26, 2025

```
[1]: !pip install --upgrade setuptools
!pip install tensorflow
!pip install tensorflow-probability
!pip install transformers
!pip install datasets
!pip install numpy
!pip install pandas
!pip install scikit-learn
!pip install matplotlib seaborn
!pip install torch torchvision torchaudio
!pip install nltk
!pip install wordcloud
!pip install torch torchvision torchaudio --index-url https://download.pytorch.
    ↳org/whl/cu121
!pip install --upgrade transformers
!pip install torch torchvision torchaudio
# Install all required libraries in one go
!pip install --upgrade pip
!pip install nltk imbalanced-learn transformers torch torchvision torchaudio
    ↳scikit-learn matplotlib seaborn wordcloud tqdm
```

Requirement already satisfied: setuptools in /opt/conda/lib/python3.10/site-packages (67.6.1)

Collecting setuptools

Downloading setuptools-80.3.1-py3-none-any.whl (1.2 MB)

1.2/1.2 MB

1.6 MB/s eta 0:00:00:00:01

Installing collected packages: setuptools

Attempting uninstall: setuptools

Found existing installation: setuptools 67.6.1

Uninstalling setuptools-67.6.1:

Successfully uninstalled setuptools-67.6.1

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

tensorflow 2.11.0 requires protobuf<3.20,>=3.9.2, but you have protobuf 3.20.3 which is incompatible.

Successfully installed setuptools-80.3.1

Requirement already satisfied: tensorflow in /opt/conda/lib/python3.10/site-packages (2.11.0)

Requirement already satisfied: wrapt>=1.11.0 in /opt/conda/lib/python3.10/site-packages (from tensorflow) (1.15.0)

Requirement already satisfied: six>=1.12.0 in /opt/conda/lib/python3.10/site-packages (from tensorflow) (1.16.0)

Requirement already satisfied: typing-extensions>=3.6.6 in /opt/conda/lib/python3.10/site-packages (from tensorflow) (4.5.0)

Requirement already satisfied: keras<2.12,>=2.11.0 in /opt/conda/lib/python3.10/site-packages (from tensorflow) (2.11.0)

Requirement already satisfied: flatbuffers>=2.0 in /opt/conda/lib/python3.10/site-packages (from tensorflow) (23.5.9)

Requirement already satisfied: tensorboard<2.12,>=2.11 in /opt/conda/lib/python3.10/site-packages (from tensorflow) (2.11.2)

Requirement already satisfied: opt-einsum>=2.3.2 in /opt/conda/lib/python3.10/site-packages (from tensorflow) (3.3.0)

Requirement already satisfied: google-pasta>=0.1.1 in /opt/conda/lib/python3.10/site-packages (from tensorflow) (0.2.0)

Collecting protobuf<3.20,>=3.9.2

Using cached

protobuf-3.19.6-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (1.1 MB)

Requirement already satisfied: gast<=0.4.0,>=0.2.1 in /opt/conda/lib/python3.10/site-packages (from tensorflow) (0.4.0)

Requirement already satisfied: packaging in /opt/conda/lib/python3.10/site-packages (from tensorflow) (23.0)

Requirement already satisfied: numpy>=1.20 in /opt/conda/lib/python3.10/site-packages (from tensorflow) (1.23.5)

Requirement already satisfied: grpcio<2.0,>=1.24.3 in /opt/conda/lib/python3.10/site-packages (from tensorflow) (1.54.2)

Requirement already satisfied: setuptools in /opt/conda/lib/python3.10/site-packages (from tensorflow) (80.3.1)

Requirement already satisfied: tensorflow-estimator<2.12,>=2.11.0 in /opt/conda/lib/python3.10/site-packages (from tensorflow) (2.11.0)

Requirement already satisfied: termcolor>=1.1.0 in /opt/conda/lib/python3.10/site-packages (from tensorflow) (2.3.0)

Requirement already satisfied: h5py>=2.9.0 in /opt/conda/lib/python3.10/site-packages (from tensorflow) (3.8.0)

Requirement already satisfied: astunparse>=1.6.0 in

/opt/conda/lib/python3.10/site-packages (from tensorflow) (1.6.3)
 Requirement already satisfied: tensorflow-io-gcs-filesystem>=0.23.1 in
 /opt/conda/lib/python3.10/site-packages (from tensorflow) (0.32.0)
 Requirement already satisfied: libclang>=13.0.0 in
 /opt/conda/lib/python3.10/site-packages (from tensorflow) (16.0.0)
 Requirement already satisfied: absl-py>=1.0.0 in /opt/conda/lib/python3.10/site-
 packages (from tensorflow) (1.4.0)
 Requirement already satisfied: wheel<1.0,>=0.23.0 in
 /opt/conda/lib/python3.10/site-packages (from astunparse>=1.6.0->tensorflow)
 (0.40.0)
 Requirement already satisfied: google-auth<3,>=1.6.3 in
 /opt/conda/lib/python3.10/site-packages (from
 tensorboard<2.12,>=2.11->tensorflow) (2.18.1)
 Requirement already satisfied: markdown>=2.6.8 in
 /opt/conda/lib/python3.10/site-packages (from
 tensorboard<2.12,>=2.11->tensorflow) (3.4.3)
 Requirement already satisfied: werkzeug>=1.0.1 in
 /opt/conda/lib/python3.10/site-packages (from
 tensorboard<2.12,>=2.11->tensorflow) (2.3.4)
 Requirement already satisfied: tensorboard-plugin-wit>=1.6.0 in
 /opt/conda/lib/python3.10/site-packages (from
 tensorboard<2.12,>=2.11->tensorflow) (1.8.1)
 Requirement already satisfied: google-auth-oauthlib<0.5,>=0.4.1 in
 /opt/conda/lib/python3.10/site-packages (from
 tensorboard<2.12,>=2.11->tensorflow) (0.4.6)
 Requirement already satisfied: tensorboard-data-server<0.7.0,>=0.6.0 in
 /opt/conda/lib/python3.10/site-packages (from
 tensorboard<2.12,>=2.11->tensorflow) (0.6.1)
 Requirement already satisfied: requests<3,>=2.21.0 in
 /opt/conda/lib/python3.10/site-packages (from
 tensorboard<2.12,>=2.11->tensorflow) (2.28.2)
 Requirement already satisfied: pyasn1-modules>=0.2.1 in
 /opt/conda/lib/python3.10/site-packages (from google-
 auth<3,>=1.6.3->tensorboard<2.12,>=2.11->tensorflow) (0.3.0)
 Requirement already satisfied: cachetools<6.0,>=2.0.0 in
 /opt/conda/lib/python3.10/site-packages (from google-
 auth<3,>=1.6.3->tensorboard<2.12,>=2.11->tensorflow) (5.3.0)
 Requirement already satisfied: urllib3<2.0 in /opt/conda/lib/python3.10/site-
 packages (from google-auth<3,>=1.6.3->tensorboard<2.12,>=2.11->tensorflow)
 (1.26.15)
 Requirement already satisfied: rsa<5,>=3.1.4 in /opt/conda/lib/python3.10/site-
 packages (from google-auth<3,>=1.6.3->tensorboard<2.12,>=2.11->tensorflow) (4.9)
 Requirement already satisfied: requests-oauthlib>=0.7.0 in
 /opt/conda/lib/python3.10/site-packages (from google-auth-
 oauthlib<0.5,>=0.4.1->tensorboard<2.12,>=2.11->tensorflow) (1.3.1)
 Requirement already satisfied: certifi>=2017.4.17 in
 /opt/conda/lib/python3.10/site-packages (from
 requests<3,>=2.21.0->tensorboard<2.12,>=2.11->tensorflow) (2023.5.7)

Requirement already satisfied: idna<4,>=2.5 in /opt/conda/lib/python3.10/site-packages (from requests<3,>=2.21.0->tensorboard<2.12,>=2.11->tensorflow) (3.4)

Requirement already satisfied: charset-normalizer<4,>=2 in /opt/conda/lib/python3.10/site-packages (from requests<3,>=2.21.0->tensorboard<2.12,>=2.11->tensorflow) (3.1.0)

Requirement already satisfied: MarkupSafe>=2.1.1 in /opt/conda/lib/python3.10/site-packages (from werkzeug>=1.0.1->tensorboard<2.12,>=2.11->tensorflow) (2.1.2)

Requirement already satisfied: pyasn1<0.6.0,>=0.4.6 in /opt/conda/lib/python3.10/site-packages (from pyasn1-modules>=0.2.1->google-auth<3,>=1.6.3->tensorboard<2.12,>=2.11->tensorflow) (0.5.0)

Requirement already satisfied: oauthlib>=3.0.0 in /opt/conda/lib/python3.10/site-packages (from requests-oauthlib>=0.7.0->google-auth-oauthlib<0.5,>=0.4.1->tensorboard<2.12,>=2.11->tensorflow) (3.2.2)

Installing collected packages: protobuf

 Attempting uninstall: protobuf

 Found existing installation: protobuf 3.20.3

 Uninstalling protobuf-3.20.3:

 Successfully uninstalled protobuf-3.20.3

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

visuldl 2.5.2 requires protobuf>=3.20.0, but you have protobuf 3.19.6 which is incompatible.

onnx 1.14.0 requires protobuf>=3.20.2, but you have protobuf 3.19.6 which is incompatible.

Successfully installed protobuf-3.19.6

Requirement already satisfied: tensorflow-probability in /opt/conda/lib/python3.10/site-packages (0.20.1)

Requirement already satisfied: absl-py in /opt/conda/lib/python3.10/site-packages (from tensorflow-probability) (1.4.0)

Requirement already satisfied: numpy>=1.13.3 in /opt/conda/lib/python3.10/site-packages (from tensorflow-probability) (1.23.5)

Requirement already satisfied: dm-tree in /opt/conda/lib/python3.10/site-packages (from tensorflow-probability) (0.1.8)

Requirement already satisfied: six>=1.10.0 in /opt/conda/lib/python3.10/site-packages (from tensorflow-probability) (1.16.0)

Requirement already satisfied: decorator in /opt/conda/lib/python3.10/site-packages (from tensorflow-probability) (5.1.1)

Requirement already satisfied: cloudpickle>=1.3 in /opt/conda/lib/python3.10/site-packages (from tensorflow-probability) (2.2.1)

Requirement already satisfied: gast>=0.3.2 in /opt/conda/lib/python3.10/site-packages (from tensorflow-probability) (0.4.0)

Collecting transformers

```

Using cached transformers-4.51.3-py3-none-any.whl (10.4 MB)
Requirement already satisfied: filelock in /opt/conda/lib/python3.10/site-
packages (from transformers) (3.12.0)
Requirement already satisfied: tqdm>=4.27 in /opt/conda/lib/python3.10/site-
packages (from transformers) (4.65.0)
Collecting huggingface-hub<1.0,>=0.30.0
  Using cached huggingface_hub-0.30.2-py3-none-any.whl (481 kB)
Collecting safetensors>=0.4.3
  Using cached
safetensors-0.5.3-cp38-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (471
kB)
Collecting tokenizers<0.22,>=0.21
  Using cached
tokenizers-0.21.1-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (3.0
MB)
Requirement already satisfied: packaging>=20.0 in
/opt/conda/lib/python3.10/site-packages (from transformers) (23.0)
Requirement already satisfied: pyyaml>=5.1 in /opt/conda/lib/python3.10/site-
packages (from transformers) (6.0)
Requirement already satisfied: requests in /opt/conda/lib/python3.10/site-
packages (from transformers) (2.28.2)
Collecting regex!=2019.12.17
  Using cached
regex-2024.11.6-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (781
kB)
Requirement already satisfied: numpy>=1.17 in /opt/conda/lib/python3.10/site-
packages (from transformers) (1.23.5)
Requirement already satisfied: typing-extensions>=3.7.4.3 in
/opt/conda/lib/python3.10/site-packages (from huggingface-
hub<1.0,>=0.30.0->transformers) (4.5.0)
Collecting fsspec>=2023.5.0
  Using cached fsspec-2025.3.2-py3-none-any.whl (194 kB)
Requirement already satisfied: urllib3<1.27,>=1.21.1 in
/opt/conda/lib/python3.10/site-packages (from requests->transformers) (1.26.15)
Requirement already satisfied: idna<4,>=2.5 in /opt/conda/lib/python3.10/site-
packages (from requests->transformers) (3.4)
Requirement already satisfied: certifi>=2017.4.17 in
/opt/conda/lib/python3.10/site-packages (from requests->transformers) (2023.5.7)
Requirement already satisfied: charset-normalizer<4,>=2 in
/opt/conda/lib/python3.10/site-packages (from requests->transformers) (3.1.0)
Installing collected packages: safetensors, regex, fsspec, huggingface-hub,
tokenizers, transformers
  Attempting uninstall: fsspec
    Found existing installation: fsspec 2023.3.0
    Uninstalling fsspec-2023.3.0:
      Successfully uninstalled fsspec-2023.3.0
Successfully installed fsspec-2025.3.2 huggingface-hub-0.30.2 regex-2024.11.6
safetensors-0.5.3 tokenizers-0.21.1 transformers-4.51.3

```

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Collecting datasets
  Using cached datasets-3.5.1-py3-none-any.whl (491 kB)
Collecting tqdm>=4.66.3
  Using cached tqdm-4.67.1-py3-none-any.whl (78 kB)
Requirement already satisfied: filelock in /opt/conda/lib/python3.10/site-
packages (from datasets) (3.12.0)
Collecting requests>=2.32.2
  Using cached requests-2.32.3-py3-none-any.whl (64 kB)
Collecting multiprocessing<0.70.17
  Using cached multiprocessing-0.70.16-py310-none-any.whl (134 kB)
Requirement already satisfied: pyyaml>=5.1 in /opt/conda/lib/python3.10/site-
packages (from datasets) (6.0)
Collecting fsspec[http]<=2025.3.0,>=2023.1.0
  Using cached fsspec-2025.3.0-py3-none-any.whl (193 kB)
Requirement already satisfied: pandas in /opt/conda/lib/python3.10/site-packages
(from datasets) (2.0.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in
/opt/conda/lib/python3.10/site-packages (from datasets) (0.3.6)
Requirement already satisfied: huggingface-hub>=0.24.0 in
/opt/conda/lib/python3.10/site-packages (from datasets) (0.30.2)
Requirement already satisfied: packaging in /opt/conda/lib/python3.10/site-
packages (from datasets) (23.0)
Requirement already satisfied: aiohttp in /opt/conda/lib/python3.10/site-
packages (from datasets) (3.8.4)
Collecting pyarrow>=15.0.0
  Using cached pyarrow-20.0.0-cp310-cp310-manylinux_2_28_x86_64.whl (42.3 MB)
Requirement already satisfied: numpy>=1.17 in /opt/conda/lib/python3.10/site-
packages (from datasets) (1.23.5)
Collecting xxhash
  Using cached
xxhash-3.5.0-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (194 kB)
Requirement already satisfied: multidict<7.0,>=4.5 in
/opt/conda/lib/python3.10/site-packages (from aiohttp->datasets) (6.0.4)
Requirement already satisfied: async-timeout<5.0,>=4.0.0a3 in
/opt/conda/lib/python3.10/site-packages (from aiohttp->datasets) (4.0.2)
Requirement already satisfied: charset-normalizer<4.0,>=2.0 in
/opt/conda/lib/python3.10/site-packages (from aiohttp->datasets) (3.1.0)
Requirement already satisfied: yarl<2.0,>=1.0 in /opt/conda/lib/python3.10/site-
packages (from aiohttp->datasets) (1.9.2)
Requirement already satisfied: frozenlist>=1.1.1 in
/opt/conda/lib/python3.10/site-packages (from aiohttp->datasets) (1.3.3)
Requirement already satisfied: attrs>=17.3.0 in /opt/conda/lib/python3.10/site-
packages (from aiohttp->datasets) (22.2.0)
Requirement already satisfied: aiosignal>=1.1.2 in
/opt/conda/lib/python3.10/site-packages (from aiohttp->datasets) (1.3.1)
Requirement already satisfied: typing-extensions>=3.7.4.3 in
/opt/conda/lib/python3.10/site-packages (from huggingface-hub>=0.24.0->datasets)
(4.5.0)

```

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Collecting dill<0.3.9,>=0.3.0
  Using cached dill-0.3.8-py3-none-any.whl (116 kB)
Requirement already satisfied: idna<4,>=2.5 in /opt/conda/lib/python3.10/site-
packages (from requests>=2.32.2->datasets) (3.4)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/opt/conda/lib/python3.10/site-packages (from requests>=2.32.2->datasets)
(1.26.15)
Requirement already satisfied: certifi>=2017.4.17 in
/opt/conda/lib/python3.10/site-packages (from requests>=2.32.2->datasets)
(2023.5.7)
Requirement already satisfied: python-dateutil>=2.8.2 in
/opt/conda/lib/python3.10/site-packages (from pandas->datasets) (2.8.2)
Requirement already satisfied: pytz>=2020.1 in /opt/conda/lib/python3.10/site-
packages (from pandas->datasets) (2023.3)
Requirement already satisfied: tzdata>=2022.1 in /opt/conda/lib/python3.10/site-
packages (from pandas->datasets) (2023.3)
Requirement already satisfied: six>=1.5 in /opt/conda/lib/python3.10/site-
packages (from python-dateutil>=2.8.2->pandas->datasets) (1.16.0)
Installing collected packages: xxhash, tqdm, requests, pyarrow, fsspec, dill,
multiprocess, datasets
  Attempting uninstall: tqdm
    Found existing installation: tqdm 4.65.0
    Uninstalling tqdm-4.65.0:
      Successfully uninstalled tqdm-4.65.0
  Attempting uninstall: requests
    Found existing installation: requests 2.28.2
    Uninstalling requests-2.28.2:
      Successfully uninstalled requests-2.28.2
  Attempting uninstall: fsspec
    Found existing installation: fsspec 2025.3.2
    Uninstalling fsspec-2025.3.2:
      Successfully uninstalled fsspec-2025.3.2
  Attempting uninstall: dill
    Found existing installation: dill 0.3.6
    Uninstalling dill-0.3.6:
      Successfully uninstalled dill-0.3.6
ERROR: pip's dependency resolver does not currently take into account all
the packages that are installed. This behaviour is the source of the following
dependency conflicts.

visuall 2.5.2 requires protobuf>=3.20.0, but you have protobuf 3.19.6 which is
incompatible.

Successfully installed datasets-3.5.1 dill-0.3.8 fsspec-2025.3.0
multiprocess-0.70.16 pyarrow-20.0.0 requests-2.32.3 tqdm-4.67.1 xxhash-3.5.0
Requirement already satisfied: numpy in /opt/conda/lib/python3.10/site-packages
(1.23.5)

```

Requirement already satisfied: pandas in /opt/conda/lib/python3.10/site-packages (2.0.0)

Requirement already satisfied: python-dateutil>=2.8.2 in /opt/conda/lib/python3.10/site-packages (from pandas) (2.8.2)

Requirement already satisfied: pytz>=2020.1 in /opt/conda/lib/python3.10/site-packages (from pandas) (2023.3)

Requirement already satisfied: tzdata>=2022.1 in /opt/conda/lib/python3.10/site-packages (from pandas) (2023.3)

Requirement already satisfied: numpy>=1.21.0 in /opt/conda/lib/python3.10/site-packages (from pandas) (1.23.5)

Requirement already satisfied: six>=1.5 in /opt/conda/lib/python3.10/site-packages (from python-dateutil>=2.8.2->pandas) (1.16.0)

Requirement already satisfied: scikit-learn in /opt/conda/lib/python3.10/site-packages (1.2.2)

Requirement already satisfied: threadpoolctl>=2.0.0 in /opt/conda/lib/python3.10/site-packages (from scikit-learn) (3.1.0)

Requirement already satisfied: scipy>=1.3.2 in /opt/conda/lib/python3.10/site-packages (from scikit-learn) (1.10.1)

Requirement already satisfied: joblib>=1.1.1 in /opt/conda/lib/python3.10/site-packages (from scikit-learn) (1.2.0)

Requirement already satisfied: numpy>=1.17.3 in /opt/conda/lib/python3.10/site-packages (from scikit-learn) (1.23.5)

Requirement already satisfied: matplotlib in /opt/conda/lib/python3.10/site-packages (3.7.1)

Requirement already satisfied: seaborn in /opt/conda/lib/python3.10/site-packages (0.12.2)

Requirement already satisfied: pillow>=6.2.0 in /opt/conda/lib/python3.10/site-packages (from matplotlib) (9.4.0)

Requirement already satisfied: pyparsing>=2.3.1 in /opt/conda/lib/python3.10/site-packages (from matplotlib) (3.0.9)

Requirement already satisfied: contourpy>=1.0.1 in /opt/conda/lib/python3.10/site-packages (from matplotlib) (1.0.7)

Requirement already satisfied: numpy>=1.20 in /opt/conda/lib/python3.10/site-packages (from matplotlib) (1.23.5)

Requirement already satisfied: cycler>=0.10 in /opt/conda/lib/python3.10/site-packages (from matplotlib) (0.11.0)

Requirement already satisfied: python-dateutil>=2.7 in /opt/conda/lib/python3.10/site-packages (from matplotlib) (2.8.2)

Requirement already satisfied: fonttools>=4.22.0 in /opt/conda/lib/python3.10/site-packages (from matplotlib) (4.39.3)

Requirement already satisfied: packaging>=20.0 in /opt/conda/lib/python3.10/site-packages (from matplotlib) (23.0)

Requirement already satisfied: kiwisolver>=1.0.1 in /opt/conda/lib/python3.10/site-packages (from matplotlib) (1.4.4)

Requirement already satisfied: pandas>=0.25 in /opt/conda/lib/python3.10/site-packages (from seaborn) (2.0.0)

Requirement already satisfied: pytz>=2020.1 in /opt/conda/lib/python3.10/site-packages (from pandas>=0.25->seaborn) (2023.3)

Requirement already satisfied: tzdata>=2022.1 in /opt/conda/lib/python3.10/site-packages (from pandas>=0.25->seaborn) (2023.3)

Requirement already satisfied: six>=1.5 in /opt/conda/lib/python3.10/site-packages (from python-dateutil>=2.7->matplotlib) (1.16.0)

Requirement already satisfied: torch in /opt/conda/lib/python3.10/site-packages (2.0.0)

Requirement already satisfied: torchvision in /opt/conda/lib/python3.10/site-packages (0.15.0)

Collecting torchaudio

Using cached torchaudio-2.7.0-cp310-cp310-manylinux_2_28_x86_64.whl (3.4 MB)

Requirement already satisfied: filelock in /opt/conda/lib/python3.10/site-packages (from torch) (3.12.0)

Requirement already satisfied: typing-extensions in /opt/conda/lib/python3.10/site-packages (from torch) (4.5.0)

Requirement already satisfied: sympy in /opt/conda/lib/python3.10/site-packages (from torch) (1.11.1)

Requirement already satisfied: networkx in /opt/conda/lib/python3.10/site-packages (from torch) (3.1)

Requirement already satisfied: Jinja2 in /opt/conda/lib/python3.10/site-packages (from torch) (3.1.2)

Requirement already satisfied: numpy in /opt/conda/lib/python3.10/site-packages (from torchvision) (1.23.5)

Requirement already satisfied: requests in /opt/conda/lib/python3.10/site-packages (from torchvision) (2.32.3)

Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in /opt/conda/lib/python3.10/site-packages (from torchvision) (9.4.0)

Collecting torch

Using cached torch-2.7.0-cp310-cp310-manylinux_2_28_x86_64.whl (865.2 MB)

Collecting typing-extensions

Using cached typing_extensions-4.13.2-py3-none-any.whl (45 kB)

Collecting nvidia-cuda-cupti-cu12==12.6.80

Using cached nvidia_cuda_cupti_cu12-12.6.80-py3-none-manylinux2014_x86_64.whl (8.9 MB)

Collecting nvidia-curand-cu12==10.3.7.77

Using cached nvidia_curand_cu12-10.3.7.77-py3-none-manylinux2014_x86_64.whl (56.3 MB)

Collecting nvidia-cudnn-cu12==9.5.1.17

Using cached nvidia_cudnn_cu12-9.5.1.17-py3-none-manylinux_2_28_x86_64.whl (571.0 MB)

Collecting nvidia-nccl-cu12==2.26.2

Using cached nvidia_nccl_cu12-2.26.2-py3-none-manylinux2014_x86_64.whl (201.3 MB)

Collecting nvidia-nvtx-cu12==12.6.77

Using cached nvidia_nvtx_cu12-12.6.77-py3-none-manylinux2014_x86_64.whl (89 kB)

Collecting nvidia-cufft-cu12==11.3.0.4

Using cached nvidia_cufft_cu12-11.3.0.4-py3-none-manylinux2014_x86_64.whl (200.2 MB)

```

Collecting nvidia-cusolver-cu12==11.7.1.2
  Using cached nvidia_cusolver_cu12-11.7.1.2-py3-none-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl (158.2 MB)
Collecting nvidia-cusparselt-cu12==0.6.3
  Using cached nvidia_cusparselt_cu12-0.6.3-py3-none-manylinux2014_x86_64.whl
(156.8 MB)
Collecting nvidia-cufile-cu12==1.11.1.6
  Using cached nvidia_cufile_cu12-1.11.1.6-py3-none-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl (1.1 MB)
Collecting nvidia-cublas-cu12==12.6.4.1
  Using cached nvidia_cublas_cu12-12.6.4.1-py3-none-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl (393.1 MB)
Requirement already satisfied: fsspec in /opt/conda/lib/python3.10/site-packages
(from torch) (2025.3.0)
Collecting nvidia-cuda-nvrtc-cu12==12.6.77
  Using cached nvidia_cuda_nvrtc_cu12-12.6.77-py3-none-manylinux2014_x86_64.whl
(23.7 MB)
Collecting triton==3.3.0
  Using cached
triton-3.3.0-cp310-cp310-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (156.4
MB)
Collecting nvidia-nvjitlink-cu12==12.6.85
  Using cached nvidia_nvjitlink_cu12-12.6.85-py3-none-
manylinux2010_x86_64.manylinux_2_12_x86_64.whl (19.7 MB)
Collecting nvidia-cuda-runtime-cu12==12.6.77
  Using cached nvidia_cuda_runtime_cu12-12.6.77-py3-none-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl (897 kB)
Collecting nvidia-cusparselt-cu12==12.5.4.2
  Using cached nvidia_cusparselt_cu12-12.5.4.2-py3-none-
manylinux2014_x86_64.manylinux_2_17_x86_64.whl (216.6 MB)
Collecting sympy
  Using cached sympy-1.14.0-py3-none-any.whl (6.3 MB)
Requirement already satisfied: setuptools>=40.8.0 in
/opt/conda/lib/python3.10/site-packages (from triton==3.3.0->torch) (80.3.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/opt/conda/lib/python3.10/site-packages (from sympy->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/opt/conda/lib/python3.10/site-packages (from jinja2->torch) (2.1.2)
Requirement already satisfied: charset-normalizer<4,>=2 in
/opt/conda/lib/python3.10/site-packages (from requests->torchvision) (3.1.0)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/opt/conda/lib/python3.10/site-packages (from requests->torchvision) (1.26.15)
Requirement already satisfied: certifi>=2017.4.17 in
/opt/conda/lib/python3.10/site-packages (from requests->torchvision) (2023.5.7)
Requirement already satisfied: idna<4,>=2.5 in /opt/conda/lib/python3.10/site-
packages (from requests->torchvision) (3.4)
Installing collected packages: nvidia-cusparselt-cu12, typing-extensions,
triton, sympy, nvidia-nvtx-cu12, nvidia-nvjitlink-cu12, nvidia-nccl-cu12,

```

nvidia-curand-cu12, nvidia-cufile-cu12, nvidia-cuda-runtime-cu12, nvidia-cuda-nvrtc-cu12, nvidia-cuda-cupti-cu12, nvidia-cublas-cu12, nvidia-cusparse-cu12, nvidia-cufft-cu12, nvidia-cudnn-cu12, nvidia-cusolver-cu12, torch, torchaudio

Attempting uninstall: typing-extensions

Found existing installation: typing_extensions 4.5.0

Uninstalling typing_extensions-4.5.0:

Successfully uninstalled typing_extensions-4.5.0

Attempting uninstall: triton

Found existing installation: triton 2.0.0

Uninstalling triton-2.0.0:

Successfully uninstalled triton-2.0.0

Attempting uninstall: sympy

Found existing installation: sympy 1.11.1

Uninstalling sympy-1.11.1:

Successfully uninstalled sympy-1.11.1

Attempting uninstall: torch

Found existing installation: torch 2.0.0

Uninstalling torch-2.0.0:

Successfully uninstalled torch-2.0.0

ERROR: pip's dependency resolver does not currently take into account all

the packages that are installed. This behaviour is the source of the following dependency conflicts.

visuldl 2.5.2 requires protobuf>=3.20.0, but you have protobuf 3.19.6 which is incompatible.

onnx 1.14.0 requires protobuf>=3.20.2, but you have protobuf 3.19.6 which is incompatible.

fastai 2.7.12 requires torch<2.1,>=1.7, but you have torch 2.7.0 which is incompatible.

Successfully installed nvidia-cublas-cu12-12.6.4.1 nvidia-cuda-cupti-cu12-12.6.80 nvidia-cuda-nvrtc-cu12-12.6.77 nvidia-cuda-runtime-cu12-12.6.77 nvidia-cudnn-cu12-9.5.1.17 nvidia-cufft-cu12-11.3.0.4 nvidia-cufile-cu12-1.11.1.6 nvidia-curand-cu12-10.3.7.77 nvidia-cusolver-cu12-11.7.1.2 nvidia-cusparse-cu12-12.5.4.2 nvidia-cusparselt-cu12-0.6.3 nvidia-nccl-cu12-2.26.2 nvidia-nvjitlink-cu12-12.6.85 nvidia-nvtx-cu12-12.6.77 sympy-1.14.0 torch-2.7.0 torchaudio-2.7.0 triton-3.3.0 typing-extensions-4.13.2

Collecting nltk

Using cached nltk-3.9.1-py3-none-any.whl (1.5 MB)

Requirement already satisfied: joblib in /opt/conda/lib/python3.10/site-packages (from nltk) (1.2.0)

Requirement already satisfied: regex>=2021.8.3 in /opt/conda/lib/python3.10/site-packages (from nltk) (2024.11.6)

Requirement already satisfied: tqdm in /opt/conda/lib/python3.10/site-packages (from nltk) (4.67.1)

Requirement already satisfied: click in /opt/conda/lib/python3.10/site-packages (from nltk) (8.1.3)

Installing collected packages: nltk

Successfully installed nltk-3.9.1

Collecting wordcloud

Using cached

wordcloud-1.9.4-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (511 kB)

Requirement already satisfied: pillow in /opt/conda/lib/python3.10/site-packages (from wordcloud) (9.4.0)

Requirement already satisfied: numpy>=1.6.1 in /opt/conda/lib/python3.10/site-packages (from wordcloud) (1.23.5)

Requirement already satisfied: matplotlib in /opt/conda/lib/python3.10/site-packages (from wordcloud) (3.7.1)

Requirement already satisfied: packaging>=20.0 in /opt/conda/lib/python3.10/site-packages (from matplotlib->wordcloud) (23.0)

Requirement already satisfied: python-dateutil>=2.7 in /opt/conda/lib/python3.10/site-packages (from matplotlib->wordcloud) (2.8.2)

Requirement already satisfied: contourpy>=1.0.1 in /opt/conda/lib/python3.10/site-packages (from matplotlib->wordcloud) (1.0.7)

Requirement already satisfied: pyparsing>=2.3.1 in /opt/conda/lib/python3.10/site-packages (from matplotlib->wordcloud) (3.0.9)

Requirement already satisfied: cycler>=0.10 in /opt/conda/lib/python3.10/site-packages (from matplotlib->wordcloud) (0.11.0)

Requirement already satisfied: kiwisolver>=1.0.1 in /opt/conda/lib/python3.10/site-packages (from matplotlib->wordcloud) (1.4.4)

Requirement already satisfied: fonttools>=4.22.0 in /opt/conda/lib/python3.10/site-packages (from matplotlib->wordcloud) (4.39.3)

Requirement already satisfied: six>=1.5 in /opt/conda/lib/python3.10/site-packages (from python-dateutil->matplotlib->wordcloud) (1.16.0)

Installing collected packages: wordcloud

Successfully installed wordcloud-1.9.4

Looking in indexes: <https://download.pytorch.org/whl/cu121>

Requirement already satisfied: torch in /opt/conda/lib/python3.10/site-packages (2.7.0)

Requirement already satisfied: torchvision in /opt/conda/lib/python3.10/site-packages (0.15.0)

Requirement already satisfied: torchaudio in /opt/conda/lib/python3.10/site-packages (2.7.0)

Requirement already satisfied: nvidia-cuda-runtime-cu12==12.6.77 in /opt/conda/lib/python3.10/site-packages (from torch) (12.6.77)

Requirement already satisfied: nvidia-cusparse-cu12==12.5.4.2 in /opt/conda/lib/python3.10/site-packages (from torch) (12.5.4.2)

Requirement already satisfied: nvidia-cufile-cu12==1.11.1.6 in /opt/conda/lib/python3.10/site-packages (from torch) (1.11.1.6)

Requirement already satisfied: typing-extensions>=4.10.0 in /opt/conda/lib/python3.10/site-packages (from torch) (4.13.2)

Requirement already satisfied: nvidia-curand-cu12==10.3.7.77 in

/opt/conda/lib/python3.10/site-packages (from torch) (10.3.7.77)
 Requirement already satisfied: nvidia-cublas-cu12==12.6.4.1 in
 /opt/conda/lib/python3.10/site-packages (from torch) (12.6.4.1)
 Requirement already satisfied: jinja2 in /opt/conda/lib/python3.10/site-packages
 (from torch) (3.1.2)
 Requirement already satisfied: nvidia-cufft-cu12==11.3.0.4 in
 /opt/conda/lib/python3.10/site-packages (from torch) (11.3.0.4)
 Requirement already satisfied: nvidia-nvjitlink-cu12==12.6.85 in
 /opt/conda/lib/python3.10/site-packages (from torch) (12.6.85)
 Requirement already satisfied: filelock in /opt/conda/lib/python3.10/site-
 packages (from torch) (3.12.0)
 Requirement already satisfied: nvidia-nvtx-cu12==12.6.77 in
 /opt/conda/lib/python3.10/site-packages (from torch) (12.6.77)
 Requirement already satisfied: nvidia-nccl-cu12==2.26.2 in
 /opt/conda/lib/python3.10/site-packages (from torch) (2.26.2)
 Requirement already satisfied: nvidia-cudnn-cu12==9.5.1.17 in
 /opt/conda/lib/python3.10/site-packages (from torch) (9.5.1.17)
 Requirement already satisfied: nvidia-cusolver-cu12==11.7.1.2 in
 /opt/conda/lib/python3.10/site-packages (from torch) (11.7.1.2)
 Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.6.77 in
 /opt/conda/lib/python3.10/site-packages (from torch) (12.6.77)
 Requirement already satisfied: nvidia-cuda-cupti-cu12==12.6.80 in
 /opt/conda/lib/python3.10/site-packages (from torch) (12.6.80)
 Requirement already satisfied: nvidia-cusparselt-cu12==0.6.3 in
 /opt/conda/lib/python3.10/site-packages (from torch) (0.6.3)
 Requirement already satisfied: fsspec in /opt/conda/lib/python3.10/site-packages
 (from torch) (2025.3.0)
 Requirement already satisfied: networkx in /opt/conda/lib/python3.10/site-
 packages (from torch) (3.1)
 Requirement already satisfied: sympy>=1.13.3 in /opt/conda/lib/python3.10/site-
 packages (from torch) (1.14.0)
 Requirement already satisfied: triton==3.3.0 in /opt/conda/lib/python3.10/site-
 packages (from torch) (3.3.0)
 Requirement already satisfied: setuptools>=40.8.0 in
 /opt/conda/lib/python3.10/site-packages (from triton==3.3.0->torch) (80.3.1)
 Requirement already satisfied: numpy in /opt/conda/lib/python3.10/site-packages
 (from torchvision) (1.23.5)
 Requirement already satisfied: requests in /opt/conda/lib/python3.10/site-
 packages (from torchvision) (2.32.3)
 Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in
 /opt/conda/lib/python3.10/site-packages (from torchvision) (9.4.0)
 Requirement already satisfied: mpmath<1.4,>=1.1.0 in
 /opt/conda/lib/python3.10/site-packages (from sympy>=1.13.3->torch) (1.3.0)
 Requirement already satisfied: MarkupSafe>=2.0 in
 /opt/conda/lib/python3.10/site-packages (from jinja2->torch) (2.1.2)
 Requirement already satisfied: charset-normalizer<4,>=2 in
 /opt/conda/lib/python3.10/site-packages (from requests->torchvision) (3.1.0)
 Requirement already satisfied: urllib3<3,>=1.21.1 in

/opt/conda/lib/python3.10/site-packages (from requests->torchvision) (1.26.15)
 Requirement already satisfied: idna<4,>=2.5 in /opt/conda/lib/python3.10/site-packages (from requests->torchvision) (3.4)
 Requirement already satisfied: certifi>=2017.4.17 in /opt/conda/lib/python3.10/site-packages (from requests->torchvision) (2023.5.7)
 Requirement already satisfied: transformers in /opt/conda/lib/python3.10/site-packages (4.51.3)
 Requirement already satisfied: huggingface-hub<1.0,>=0.30.0 in /opt/conda/lib/python3.10/site-packages (from transformers) (0.30.2)
 Requirement already satisfied: filelock in /opt/conda/lib/python3.10/site-packages (from transformers) (3.12.0)
 Requirement already satisfied: numpy>=1.17 in /opt/conda/lib/python3.10/site-packages (from transformers) (1.23.5)
 Requirement already satisfied: requests in /opt/conda/lib/python3.10/site-packages (from transformers) (2.32.3)
 Requirement already satisfied: tokenizers<0.22,>=0.21 in /opt/conda/lib/python3.10/site-packages (from transformers) (0.21.1)
 Requirement already satisfied: packaging>=20.0 in /opt/conda/lib/python3.10/site-packages (from transformers) (23.0)
 Requirement already satisfied: safetensors>=0.4.3 in /opt/conda/lib/python3.10/site-packages (from transformers) (0.5.3)
 Requirement already satisfied: regex!=2019.12.17 in /opt/conda/lib/python3.10/site-packages (from transformers) (2024.11.6)
 Requirement already satisfied: pyyaml>=5.1 in /opt/conda/lib/python3.10/site-packages (from transformers) (6.0)
 Requirement already satisfied: tqdm>=4.27 in /opt/conda/lib/python3.10/site-packages (from transformers) (4.67.1)
 Requirement already satisfied: typing-extensions>=3.7.4.3 in /opt/conda/lib/python3.10/site-packages (from huggingface-hub<1.0,>=0.30.0->transformers) (4.13.2)
 Requirement already satisfied: fsspec>=2023.5.0 in /opt/conda/lib/python3.10/site-packages (from huggingface-hub<1.0,>=0.30.0->transformers) (2025.3.0)
 Requirement already satisfied: certifi>=2017.4.17 in /opt/conda/lib/python3.10/site-packages (from requests->transformers) (2023.5.7)
 Requirement already satisfied: urllib3<3,>=1.21.1 in /opt/conda/lib/python3.10/site-packages (from requests->transformers) (1.26.15)
 Requirement already satisfied: charset-normalizer<4,>=2 in /opt/conda/lib/python3.10/site-packages (from requests->transformers) (3.1.0)
 Requirement already satisfied: idna<4,>=2.5 in /opt/conda/lib/python3.10/site-packages (from requests->transformers) (3.4)
 Requirement already satisfied: torch in /opt/conda/lib/python3.10/site-packages (2.7.0)
 Requirement already satisfied: torchvision in /opt/conda/lib/python3.10/site-packages (0.15.0)
 Requirement already satisfied: torchaudio in /opt/conda/lib/python3.10/site-packages (2.7.0)
 Requirement already satisfied: nvidia-nccl-cu12==2.26.2 in

/opt/conda/lib/python3.10/site-packages (from torch) (2.26.2)
 Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.6.77 in
 /opt/conda/lib/python3.10/site-packages (from torch) (12.6.77)
 Requirement already satisfied: nvidia-cusparse-cu12==12.5.4.2 in
 /opt/conda/lib/python3.10/site-packages (from torch) (12.5.4.2)
 Requirement already satisfied: sympy>=1.13.3 in /opt/conda/lib/python3.10/site-
 packages (from torch) (1.14.0)
 Requirement already satisfied: nvidia-cublas-cu12==12.6.4.1 in
 /opt/conda/lib/python3.10/site-packages (from torch) (12.6.4.1)
 Requirement already satisfied: nvidia-cudnn-cu12==9.5.1.17 in
 /opt/conda/lib/python3.10/site-packages (from torch) (9.5.1.17)
 Requirement already satisfied: nvidia-cusolver-cu12==11.7.1.2 in
 /opt/conda/lib/python3.10/site-packages (from torch) (11.7.1.2)
 Requirement already satisfied: nvidia-cusparselt-cu12==0.6.3 in
 /opt/conda/lib/python3.10/site-packages (from torch) (0.6.3)
 Requirement already satisfied: nvidia-cuda-cupti-cu12==12.6.80 in
 /opt/conda/lib/python3.10/site-packages (from torch) (12.6.80)
 Requirement already satisfied: nvidia-cuda-runtime-cu12==12.6.77 in
 /opt/conda/lib/python3.10/site-packages (from torch) (12.6.77)
 Requirement already satisfied: typing-extensions>=4.10.0 in
 /opt/conda/lib/python3.10/site-packages (from torch) (4.13.2)
 Requirement already satisfied: nvidia-nvtx-cu12==12.6.77 in
 /opt/conda/lib/python3.10/site-packages (from torch) (12.6.77)
 Requirement already satisfied: nvidia-curand-cu12==10.3.7.77 in
 /opt/conda/lib/python3.10/site-packages (from torch) (10.3.7.77)
 Requirement already satisfied: jinja2 in /opt/conda/lib/python3.10/site-packages
 (from torch) (3.1.2)
 Requirement already satisfied: networkx in /opt/conda/lib/python3.10/site-
 packages (from torch) (3.1)
 Requirement already satisfied: triton==3.3.0 in /opt/conda/lib/python3.10/site-
 packages (from torch) (3.3.0)
 Requirement already satisfied: filelock in /opt/conda/lib/python3.10/site-
 packages (from torch) (3.12.0)
 Requirement already satisfied: nvidia-cufile-cu12==1.11.1.6 in
 /opt/conda/lib/python3.10/site-packages (from torch) (1.11.1.6)
 Requirement already satisfied: nvidia-cufft-cu12==11.3.0.4 in
 /opt/conda/lib/python3.10/site-packages (from torch) (11.3.0.4)
 Requirement already satisfied: fsspec in /opt/conda/lib/python3.10/site-packages
 (from torch) (2025.3.0)
 Requirement already satisfied: nvidia-nvjitlink-cu12==12.6.85 in
 /opt/conda/lib/python3.10/site-packages (from torch) (12.6.85)
 Requirement already satisfied: setuptools>=40.8.0 in
 /opt/conda/lib/python3.10/site-packages (from triton==3.3.0->torch) (80.3.1)
 Requirement already satisfied: numpy in /opt/conda/lib/python3.10/site-packages
 (from torchvision) (1.23.5)
 Requirement already satisfied: requests in /opt/conda/lib/python3.10/site-
 packages (from torchvision) (2.32.3)
 Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in

/opt/conda/lib/python3.10/site-packages (from torchvision) (9.4.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/opt/conda/lib/python3.10/site-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/opt/conda/lib/python3.10/site-packages (from jinja2->torch) (2.1.2)
Requirement already satisfied: charset-normalizer<4,>=2 in
/opt/conda/lib/python3.10/site-packages (from requests->torchvision) (3.1.0)
Requirement already satisfied: idna<4,>=2.5 in /opt/conda/lib/python3.10/site-
packages (from requests->torchvision) (3.4)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/opt/conda/lib/python3.10/site-packages (from requests->torchvision) (1.26.15)
Requirement already satisfied: certifi>=2017.4.17 in
/opt/conda/lib/python3.10/site-packages (from requests->torchvision) (2023.5.7)
Requirement already satisfied: pip in /opt/conda/lib/python3.10/site-packages
(23.0.1)

Collecting pip

Using cached pip-25.1.1-py3-none-any.whl (1.8 MB)

Installing collected packages: pip

Attempting uninstall: pip

Found existing installation: pip 23.0.1

Uninstalling pip-23.0.1:

Successfully uninstalled pip-23.0.1

ERROR: pip's dependency resolver does not currently take into account all
the packages that are installed. This behaviour is the source of the following
dependency conflicts.

fastai 2.7.12 requires torch<2.1,>=1.7, but you have torch 2.7.0 which is
incompatible.

Successfully installed pip-25.1.1

Requirement already satisfied: nltk in /opt/conda/lib/python3.10/site-packages
(3.9.1)

Collecting imbalanced-learn

Using cached imbalanced_learn-0.13.0-py3-none-any.whl.metadata (8.8 kB)

Requirement already satisfied: transformers in /opt/conda/lib/python3.10/site-
packages (4.51.3)

Requirement already satisfied: torch in /opt/conda/lib/python3.10/site-packages
(2.7.0)

Requirement already satisfied: torchvision in /opt/conda/lib/python3.10/site-
packages (0.15.0)

Requirement already satisfied: torchaudio in /opt/conda/lib/python3.10/site-
packages (2.7.0)

Requirement already satisfied: scikit-learn in /opt/conda/lib/python3.10/site-
packages (1.2.2)

Requirement already satisfied: matplotlib in /opt/conda/lib/python3.10/site-
packages (3.7.1)

Requirement already satisfied: seaborn in /opt/conda/lib/python3.10/site-

packages (0.12.2)
 Requirement already satisfied: wordcloud in /opt/conda/lib/python3.10/site-packages (1.9.4)
 Requirement already satisfied: tqdm in /opt/conda/lib/python3.10/site-packages (4.67.1)
 Requirement already satisfied: click in /opt/conda/lib/python3.10/site-packages (from nltk) (8.1.3)
 Requirement already satisfied: joblib in /opt/conda/lib/python3.10/site-packages (from nltk) (1.2.0)
 Requirement already satisfied: regex<2021.8.3 in /opt/conda/lib/python3.10/site-packages (from nltk) (2024.11.6)
 Collecting numpy<3,>=1.24.3 (from imbalanced-learn)
 Using cached
 numpy-2.2.5-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (62 kB)
 Requirement already satisfied: scipy<2,>=1.10.1 in /opt/conda/lib/python3.10/site-packages (from imbalanced-learn) (1.10.1)
 Collecting scikit-learn
 Using cached scikit_learn-1.6.1-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (18 kB)
 Collecting sklearn-compat<1,>=0.1 (from imbalanced-learn)
 Using cached sklearn_compat-0.1.3-py3-none-any.whl.metadata (18 kB)
 Requirement already satisfied: threadpoolctl<4,>=2.0.0 in /opt/conda/lib/python3.10/site-packages (from imbalanced-learn) (3.1.0)
 Collecting numpy<3,>=1.24.3 (from imbalanced-learn)
 Using cached
 numpy-1.26.4-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (61 kB)
 Requirement already satisfied: filelock in /opt/conda/lib/python3.10/site-packages (from transformers) (3.12.0)
 Requirement already satisfied: huggingface-hub<1.0,>=0.30.0 in /opt/conda/lib/python3.10/site-packages (from transformers) (0.30.2)
 Requirement already satisfied: packaging>=20.0 in /opt/conda/lib/python3.10/site-packages (from transformers) (23.0)
 Requirement already satisfied: pyyaml>=5.1 in /opt/conda/lib/python3.10/site-packages (from transformers) (6.0)
 Requirement already satisfied: requests in /opt/conda/lib/python3.10/site-packages (from transformers) (2.32.3)
 Requirement already satisfied: tokenizers<0.22,>=0.21 in /opt/conda/lib/python3.10/site-packages (from transformers) (0.21.1)
 Requirement already satisfied: safetensors>=0.4.3 in /opt/conda/lib/python3.10/site-packages (from transformers) (0.5.3)
 Requirement already satisfied: fsspec>=2023.5.0 in /opt/conda/lib/python3.10/site-packages (from huggingface-hub<1.0,>=0.30.0->transformers) (2025.3.0)
 Requirement already satisfied: typing-extensions>=3.7.4.3 in /opt/conda/lib/python3.10/site-packages (from huggingface-hub<1.0,>=0.30.0->transformers) (4.13.2)

Requirement already satisfied: sympy>=1.13.3 in /opt/conda/lib/python3.10/site-packages (from torch) (1.14.0)

Requirement already satisfied: networkx in /opt/conda/lib/python3.10/site-packages (from torch) (3.1)

Requirement already satisfied: jinja2 in /opt/conda/lib/python3.10/site-packages (from torch) (3.1.2)

Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.6.77 in /opt/conda/lib/python3.10/site-packages (from torch) (12.6.77)

Requirement already satisfied: nvidia-cuda-runtime-cu12==12.6.77 in /opt/conda/lib/python3.10/site-packages (from torch) (12.6.77)

Requirement already satisfied: nvidia-cuda-cupti-cu12==12.6.80 in /opt/conda/lib/python3.10/site-packages (from torch) (12.6.80)

Requirement already satisfied: nvidia-cudnn-cu12==9.5.1.17 in /opt/conda/lib/python3.10/site-packages (from torch) (9.5.1.17)

Requirement already satisfied: nvidia-cublas-cu12==12.6.4.1 in /opt/conda/lib/python3.10/site-packages (from torch) (12.6.4.1)

Requirement already satisfied: nvidia-cufft-cu12==11.3.0.4 in /opt/conda/lib/python3.10/site-packages (from torch) (11.3.0.4)

Requirement already satisfied: nvidia-curand-cu12==10.3.7.77 in /opt/conda/lib/python3.10/site-packages (from torch) (10.3.7.77)

Requirement already satisfied: nvidia-cusolver-cu12==11.7.1.2 in /opt/conda/lib/python3.10/site-packages (from torch) (11.7.1.2)

Requirement already satisfied: nvidia-cuspars-cu12==12.5.4.2 in /opt/conda/lib/python3.10/site-packages (from torch) (12.5.4.2)

Requirement already satisfied: nvidia-cusparselt-cu12==0.6.3 in /opt/conda/lib/python3.10/site-packages (from torch) (0.6.3)

Requirement already satisfied: nvidia-nccl-cu12==2.26.2 in /opt/conda/lib/python3.10/site-packages (from torch) (2.26.2)

Requirement already satisfied: nvidia-nvtx-cu12==12.6.77 in /opt/conda/lib/python3.10/site-packages (from torch) (12.6.77)

Requirement already satisfied: nvidia-nvjitlink-cu12==12.6.85 in /opt/conda/lib/python3.10/site-packages (from torch) (12.6.85)

Requirement already satisfied: nvidia-cufile-cu12==1.11.1.6 in /opt/conda/lib/python3.10/site-packages (from torch) (1.11.1.6)

Requirement already satisfied: triton==3.3.0 in /opt/conda/lib/python3.10/site-packages (from torch) (3.3.0)

Requirement already satisfied: setuptools>=40.8.0 in /opt/conda/lib/python3.10/site-packages (from triton==3.3.0->torch) (80.3.1)

Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in /opt/conda/lib/python3.10/site-packages (from torchvision) (9.4.0)

Requirement already satisfied: contourpy>=1.0.1 in /opt/conda/lib/python3.10/site-packages (from matplotlib) (1.0.7)

Requirement already satisfied: cycler>=0.10 in /opt/conda/lib/python3.10/site-packages (from matplotlib) (0.11.0)

Requirement already satisfied: fonttools>=4.22.0 in /opt/conda/lib/python3.10/site-packages (from matplotlib) (4.39.3)

Requirement already satisfied: kiwisolver>=1.0.1 in /opt/conda/lib/python3.10/site-packages (from matplotlib) (1.4.4)

```

Requirement already satisfied: pyparsing>=2.3.1 in
/opt/conda/lib/python3.10/site-packages (from matplotlib) (3.0.9)
Requirement already satisfied: python-dateutil>=2.7 in
/opt/conda/lib/python3.10/site-packages (from matplotlib) (2.8.2)
Requirement already satisfied: pandas>=0.25 in /opt/conda/lib/python3.10/site-
packages (from seaborn) (2.0.0)
Requirement already satisfied: pytz>=2020.1 in /opt/conda/lib/python3.10/site-
packages (from pandas>=0.25->seaborn) (2023.3)
Requirement already satisfied: tzdata>=2022.1 in /opt/conda/lib/python3.10/site-
packages (from pandas>=0.25->seaborn) (2023.3)
Requirement already satisfied: six>=1.5 in /opt/conda/lib/python3.10/site-
packages (from python-dateutil>=2.7->matplotlib) (1.16.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/opt/conda/lib/python3.10/site-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/opt/conda/lib/python3.10/site-packages (from jinja2->torch) (2.1.2)
Requirement already satisfied: charset-normalizer<4,>=2 in
/opt/conda/lib/python3.10/site-packages (from requests->transformers) (3.1.0)
Requirement already satisfied: idna<4,>=2.5 in /opt/conda/lib/python3.10/site-
packages (from requests->transformers) (3.4)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/opt/conda/lib/python3.10/site-packages (from requests->transformers) (1.26.15)
Requirement already satisfied: certifi>=2017.4.17 in
/opt/conda/lib/python3.10/site-packages (from requests->transformers) (2023.5.7)
Using cached imbalanced_learn-0.13.0-py3-none-any.whl (238 kB)
Using cached
scikit_learn-1.6.1-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl
(13.5 MB)
Using cached
numpy-1.26.4-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (18.2
MB)
Using cached sklearn_compat-0.1.3-py3-none-any.whl (18 kB)
Installing collected packages: numpy, scikit-learn, sklearn-compat, imbalanced-
learn
  Attempting uninstall: numpy
    Found existing installation: numpy 1.23.5
    Uninstalling numpy-1.23.5:
      Successfully uninstalled numpy-1.23.5    0/4 [numpy]
  Attempting uninstall: scikit-learn    0/4 [numpy]
    Found existing installation: scikit-learn 1.2.2
    Uninstalling scikit-learn-1.2.2:
      Successfully uninstalled scikit-learn-1.2.2
1/4 [scikit-learn]
4/4
[imbalanced-learn][imbalanced-learn]

```

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

fastai 2.7.12 requires torch<2.1,>=1.7, but you have torch 2.7.0 which is incompatible.

numba 0.56.4 requires numpy<1.24,>=1.18, but you have numpy 1.26.4 which is incompatible.

onnx 1.14.0 requires protobuf>=3.20.2, but you have protobuf 3.19.6 which is incompatible.

visualdl 2.5.2 requires protobuf>=3.20.0, but you have protobuf 3.19.6 which is incompatible.

Successfully installed imbalanced-learn-0.13.0 numpy-1.26.4 scikit-learn-1.6.1 sklearn-compat-0.1.3

```
[2]: # 1) IMPORT ALL REQUIRED LIBRARIES

import pandas as pd
import numpy as np
import re
import nltk
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
from nltk.sentiment.vader import SentimentIntensityAnalyzer

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, accuracy_score, \
    confusion_matrix
from sklearn.utils import compute_class_weight

from imblearn.over_sampling import RandomOverSampler

import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader

from transformers import (
    BertTokenizer,
    BertModel,
    BertForSequenceClassification,
```

```

        get_scheduler
    )
    from torch.optim import AdamW
    import matplotlib.pyplot as plt
    import seaborn as sns
    from wordcloud import WordCloud
    from tqdm.auto import tqdm

    nltk.download('stopwords')
    nltk.download('wordnet')
    nltk.download('omw-1.4')
    nltk.download('vader_lexicon')

    # 2) LOAD THE DATASET
    file_path = "/home/jovyan/Project/Combined Data.csv"
    df = pd.read_csv(file_path)

```

```

/opt/conda/lib/python3.10/site-packages/torchvision/io/image.py:13: UserWarning:
Failed to load image Python extension: '/opt/conda/lib/python3.10/site-
packages/torchvision/image.so: undefined symbol:
_ZN3c1017RegisterOperatorsD1Ev'If you don't plan on using image functionality
from `torchvision.io`, you can ignore this warning. Otherwise, there might be
something wrong with your environment. Did you have `libjpeg` or `libpng`
installed before building `torchvision` from source?
  warn(
2025-05-07 02:04:43.933673: I tensorflow/core/platform/cpu_feature_guard.cc:193]
This TensorFlow binary is optimized with oneAPI Deep Neural Network Library
(oneDNN) to use the following CPU instructions in performance-critical
operations:  AVX2 FMA
To enable them in other operations, rebuild TensorFlow with the appropriate
compiler flags.
2025-05-07 02:04:44.547430: W
tensorflow/compiler/xla/stream_executor/platform/default/dso_loader.cc:64] Could
not load dynamic library 'libnvinfer.so.7'; dLError: libnvinfer.so.7: cannot
open shared object file: No such file or directory; LD_LIBRARY_PATH:
/usr/local/nvidia/lib:/usr/local/nvidia/lib64
2025-05-07 02:04:44.547534: W
tensorflow/compiler/xla/stream_executor/platform/default/dso_loader.cc:64] Could
not load dynamic library 'libnvinfer_plugin.so.7'; dLError:
libnvinfer_plugin.so.7: cannot open shared object file: No such file or
directory; LD_LIBRARY_PATH: /usr/local/nvidia/lib:/usr/local/nvidia/lib64
2025-05-07 02:04:44.547544: W
tensorflow/compiler/tf2tensorrt/utils/py_utils.cc:38] TF-TRT Warning: Cannot
dlopen some TensorRT libraries. If you would like to use Nvidia GPU with
TensorRT, please make sure the missing libraries mentioned above are installed
properly.
[nltk_data] Downloading package stopwords to /home/jovyan/nltk_data...

```

```
[nltk_data] Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to /home/jovyan/nltk_data...
[nltk_data] Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to /home/jovyan/nltk_data...
[nltk_data] Package omw-1.4 is already up-to-date!
[nltk_data] Downloading package vader_lexicon to
[nltk_data] /home/jovyan/nltk_data...
[nltk_data] Package vader_lexicon is already up-to-date!
```

[3]: *# 3) SHOW THE STRUCTURE OF DATASET*

```
print(df.head())
print(df.info())
print(df.describe(include='all'))
```

```

      Unnamed: 0      statement      status
0             0             oh my gosh  Anxiety
1             1  trouble sleeping, confused mind, restless hear...  Anxiety
2             2  All wrong, back off dear, forward doubt. Stay ...  Anxiety
3             3  I've shifted my focus to something else but I'...  Anxiety
4             4  I'm restless and restless, it's been a month n...  Anxiety

```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 53043 entries, 0 to 53042
```

```
Data columns (total 3 columns):
```

#	Column	Non-Null Count	Dtype
0	Unnamed: 0	53043 non-null	int64
1	statement	52681 non-null	object
2	status	53043 non-null	object

```
dtypes: int64(1), object(2)
```

```
memory usage: 1.2+ MB
```

```
None
```

	Unnamed: 0	statement	status
count	53043.000000	52681	53043
unique	NaN	51073	7
top	NaN	what do you mean?	Normal
freq	NaN	22	16351
mean	26521.000000	NaN	NaN
std	15312.339501	NaN	NaN
min	0.000000	NaN	NaN
25%	13260.500000	NaN	NaN
50%	26521.000000	NaN	NaN
75%	39781.500000	NaN	NaN
max	53042.000000	NaN	NaN

[4]: *# 4) CLEAN THE DATASET*

```
df = df.loc[:, ~df.columns.str.contains('~Unnamed')]
df.drop_duplicates(inplace=True)
```

```

df.dropna(subset=['statement', 'status'], inplace=True)

lemmatizer = WordNetLemmatizer()
stops = set(stopwords.words('english'))

def clean_text(text):
    text = text.lower()
    text = re.sub(r'[^a-z\s]', ' ', text)
    text = re.sub(r'\s+', ' ', text).strip()
    tokens = [w for w in text.split() if w not in stops]
    return ' '.join(lemmatizer.lemmatize(w) for w in tokens)

df['statement'] = df['statement'].apply(clean_text)

sia = SentimentIntensityAnalyzer()
df['sentiment_score'] = df['statement'].apply(lambda t: sia.
    ↪polarity_scores(t)['compound'])

```

```

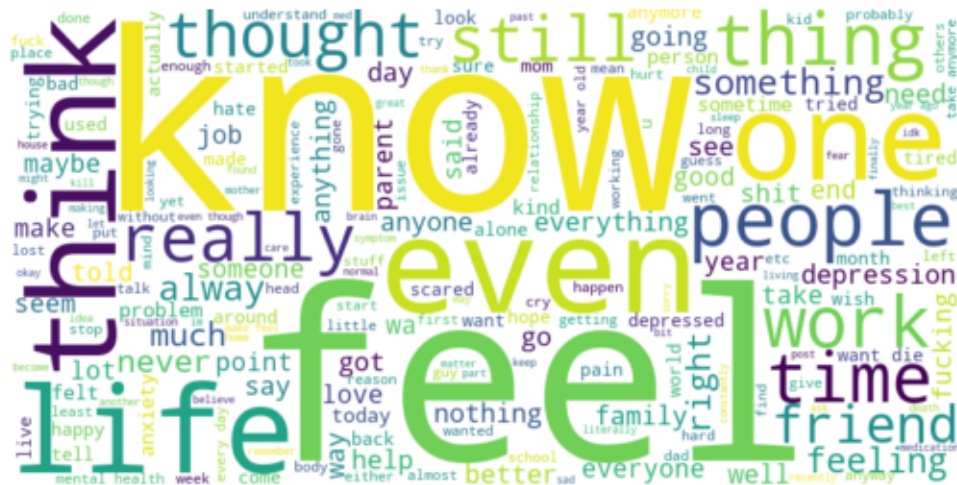
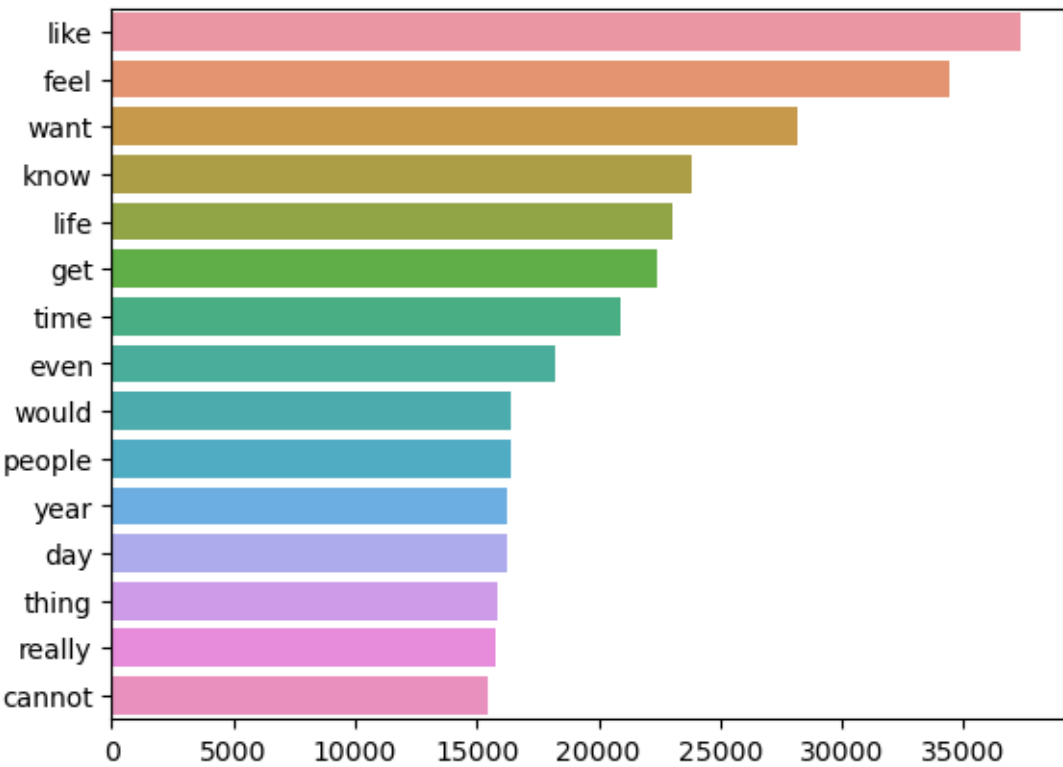
[5]: # 5) TEXT PROCESSING AND VISUALIZATION
all_words = ' '.join(df['statement']).split()
freq = pd.Series(all_words).value_counts().head(15)
sns.barplot(x=freq.values, y=freq.index)
plt.title("Top 15 Words"); plt.show()

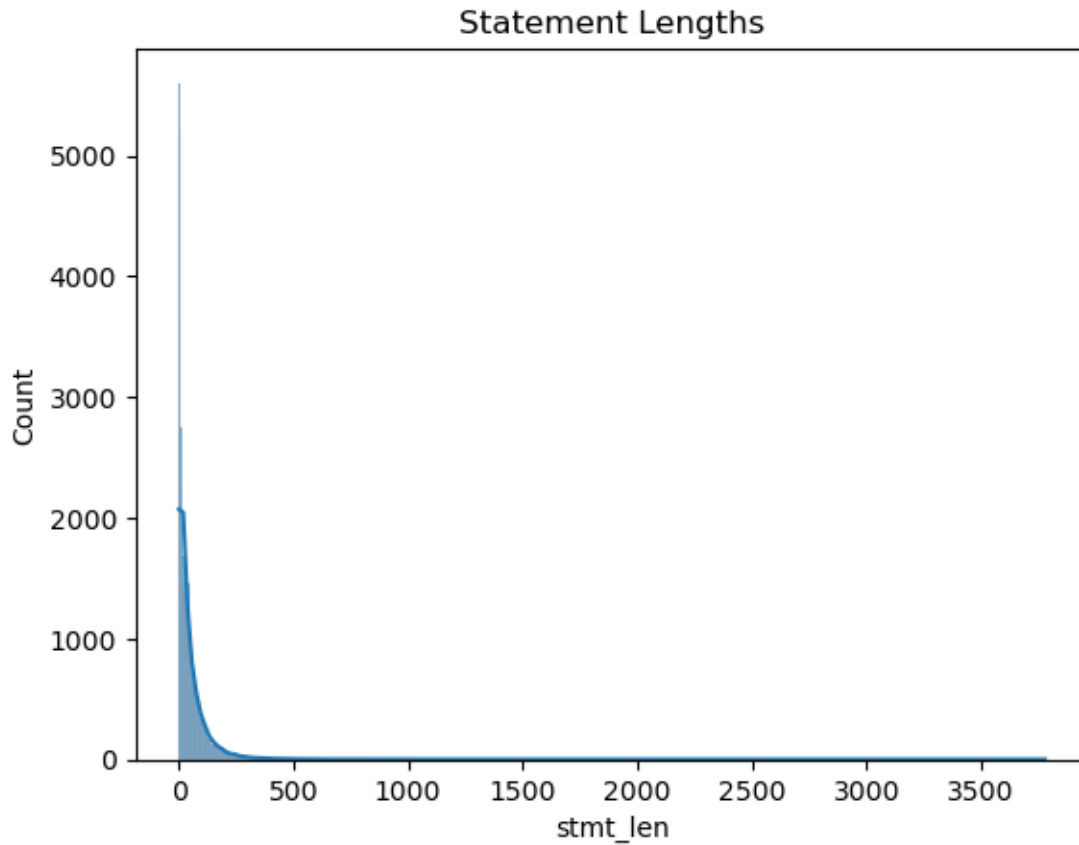
wc = WordCloud(width=800, height=400, background_color='white').generate(' '.
    ↪join(df['statement']))
plt.imshow(wc, interpolation='bilinear'); plt.axis('off'); plt.show()

df['stmt_len'] = df['statement'].apply(lambda s: len(s.split()))
sns.histplot(df['stmt_len'], kde=True)
plt.title("Statement Lengths"); plt.show()

```

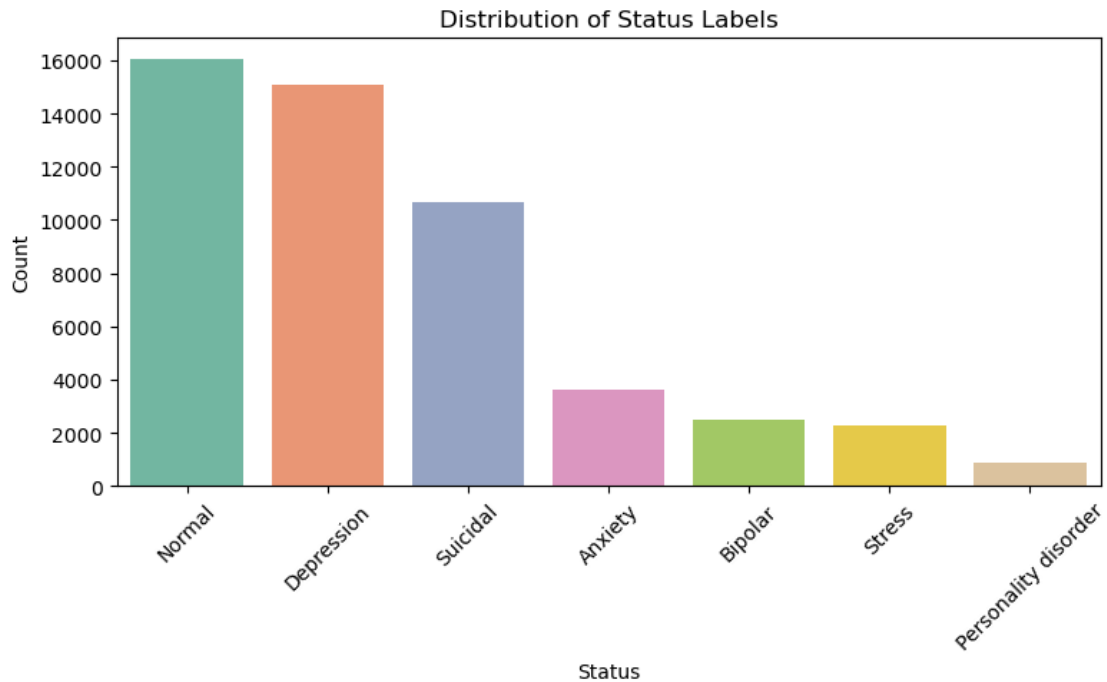
Top 15 Words





```
[6]: import seaborn as sns
import matplotlib.pyplot as plt

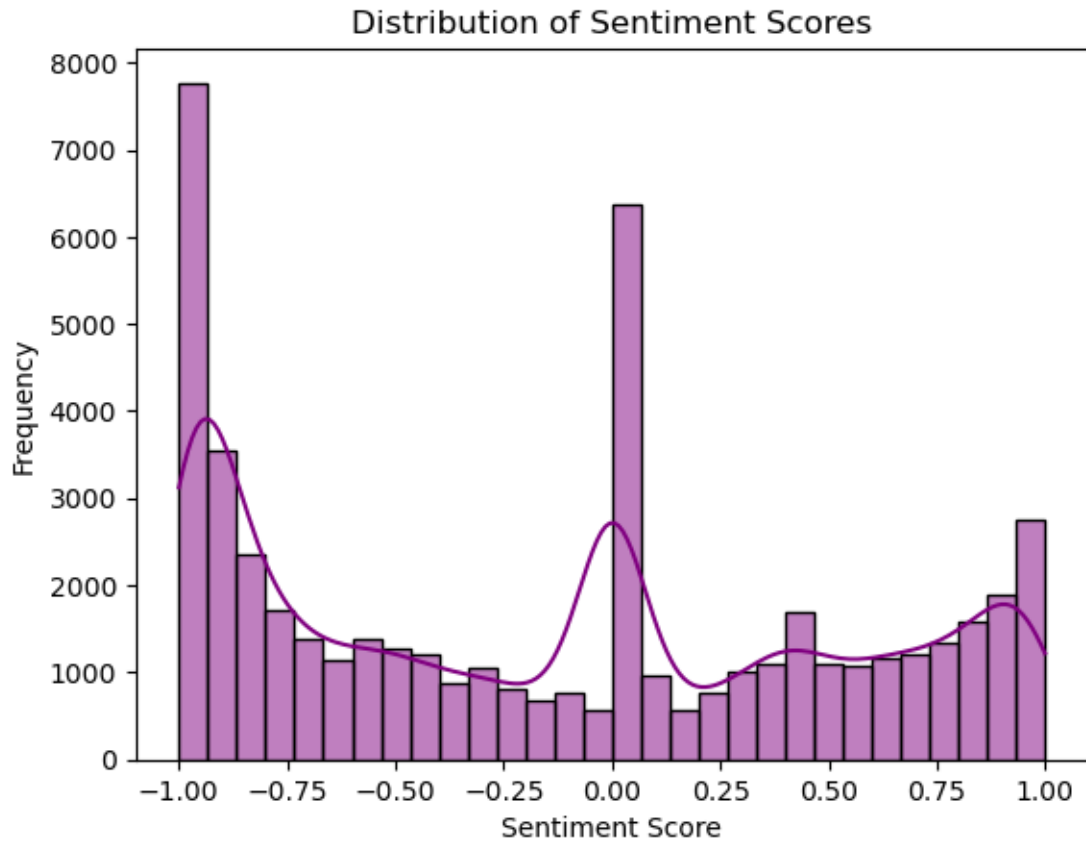
plt.figure(figsize=(8, 5))
sns.countplot(data=df, x='status', order=df['status'].value_counts().index,
             palette='Set2')
plt.title("Distribution of Status Labels")
plt.xlabel("Status")
plt.ylabel("Count")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



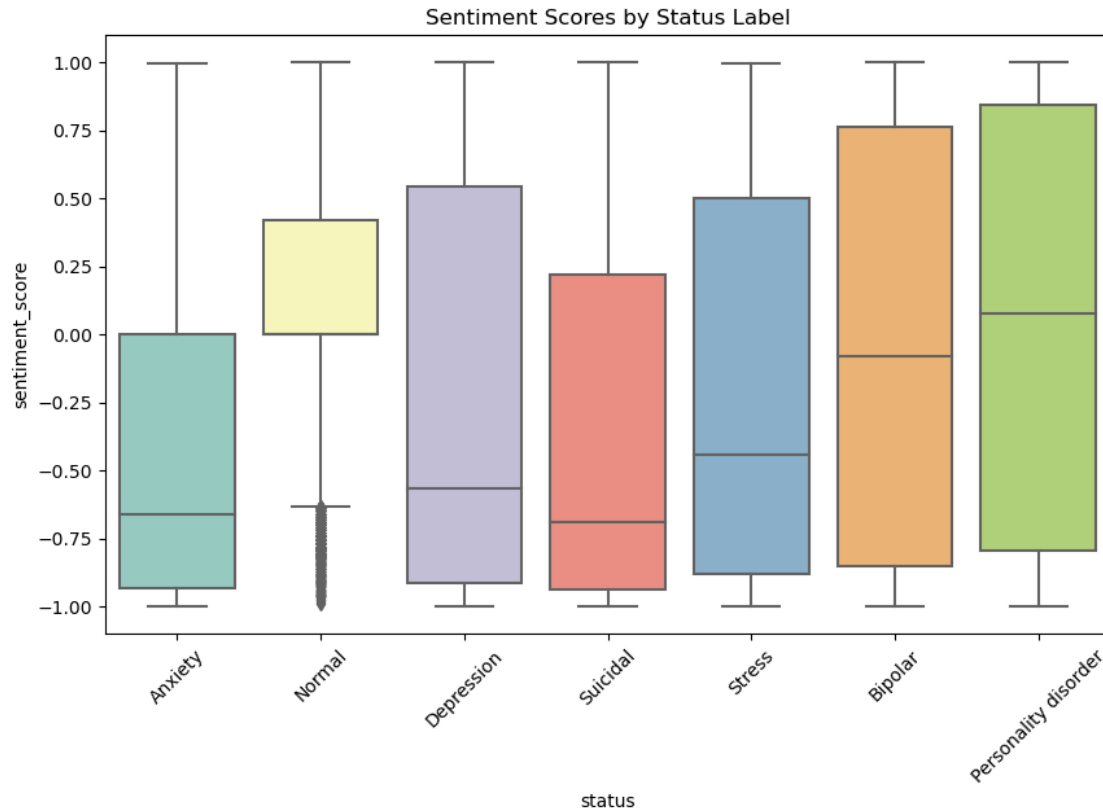
```
[7]: status_percent = df['status'].value_counts(normalize=True) * 100
print("Status distribution (percent):")
print(status_percent.round(2))
```

```
Status distribution (percent):
status
Normal          31.39
Depression       29.54
Suicidal         20.83
Anxiety          7.09
Bipolar          4.89
Stress           4.49
Personality disorder 1.75
Name: proportion, dtype: float64
```

```
[8]: sns.histplot(df['sentiment_score'], kde=True, bins=30, color='purple')
plt.title("Distribution of Sentiment Scores")
plt.xlabel("Sentiment Score")
plt.ylabel("Frequency")
plt.show()
```



```
[9]: plt.figure(figsize=(10, 6))
sns.boxplot(x='status', y='sentiment_score', data=df, palette='Set3')
plt.title("Sentiment Scores by Status Label")
plt.xticks(rotation=45)
plt.show()
#entiment scores by status
```



```
[10]: # Before vs After Cleaning Examples
sample_df = df[['statement']].copy()
sample_df['original'] = pd.read_csv(file_path)['statement']
sample_df = sample_df.dropna().head(10)
print(sample_df[['original', 'statement']])
```

	original
0	oh my gosh \
1	trouble sleeping, confused mind, restless hear...
2	All wrong, back off dear, forward doubt. Stay ...
3	I've shifted my focus to something else but I'...
4	I'm restless and restless, it's been a month n...
5	every break, you must be nervous, like somethi...
6	I feel scared, anxious, what can I do? And may...
7	Have you ever felt nervous but didn't know why?
8	I haven't slept well for 2 days, it's like I'm...
9	I'm really worried, I want to cry.

	statement
0	oh gosh
1	trouble sleeping confused mind restless heart ...

```

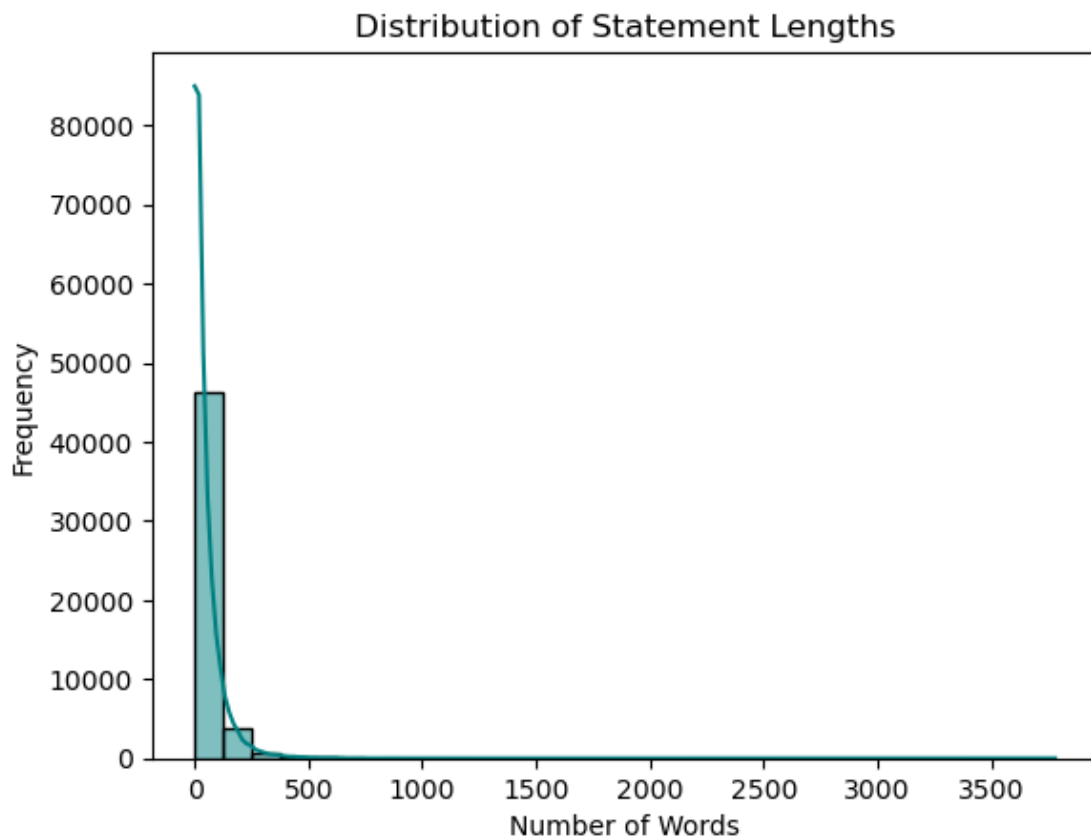
2 wrong back dear forward doubt stay restless re...
3     shifted focus something else still worried
4         restless restless month boy mean
5 every break must nervous like something wrong ...
6     feel scared anxious may family u protected
7         ever felt nervous know
8         slept well day like restless huh
9         really worried want cry

```

```

[11]: df['stmt_len'] = df['statement'].apply(lambda s: len(s.split()))
sns.histplot(df['stmt_len'], bins=30, kde=True, color='teal')
plt.title("Distribution of Statement Lengths")
plt.xlabel("Number of Words")
plt.ylabel("Frequency")
plt.show()

```



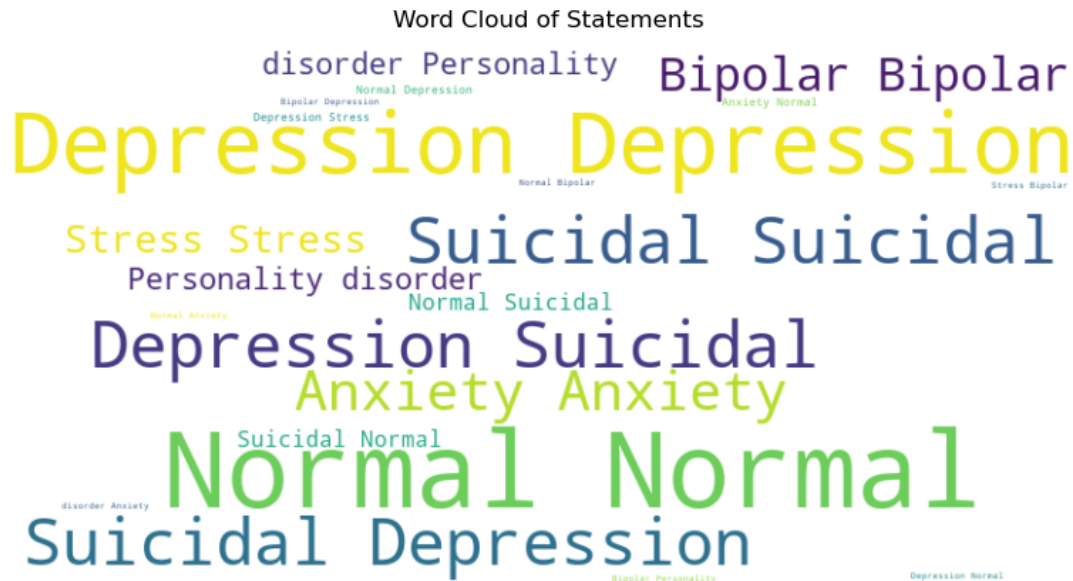
```

[12]: from wordcloud import WordCloud

text = ' '.join(df['status'])
wc = WordCloud(width=800, height=400, background_color='white').generate(text)

```

```
plt.figure(figsize=(10, 5))
plt.imshow(wc, interpolation='bilinear')
plt.axis('off')
plt.title("Word Cloud of Statements")
plt.show()
```



```
[13]: # 6) LABEL ENCODING
le = LabelEncoder()
df['status_enc'] = le.fit_transform(df['status'])
```

```
[14]: # 7) SPLIT + OVERSAMPLING

X_train, X_test, y_train, y_test = train_test_split(
    df[['statement', 'status', 'sentiment_score', 'stmt_len']],
    df['status_enc'],
    test_size=0.2,
    stratify=df['status_enc'],
    random_state=42
)
```

```
[17]: !pip install textblob

from textblob import TextBlob

# Define backtranslation
def back_translate(text):
```

```

try:
    return str(TextBlob(text).translate(to='fr').translate(to='en'))
except Exception:
    return text

```

Collecting textblob

```

Downloading textblob-0.19.0-py3-none-any.whl.metadata (4.4 kB)
Requirement already satisfied: nltk>=3.9 in /opt/conda/lib/python3.10/site-
packages (from textblob) (3.9.1)
Requirement already satisfied: click in /opt/conda/lib/python3.10/site-packages
(from nltk>=3.9->textblob) (8.1.3)
Requirement already satisfied: joblib in /opt/conda/lib/python3.10/site-packages
(from nltk>=3.9->textblob) (1.2.0)
Requirement already satisfied: regex>=2021.8.3 in
/opt/conda/lib/python3.10/site-packages (from nltk>=3.9->textblob) (2024.11.6)
Requirement already satisfied: tqdm in /opt/conda/lib/python3.10/site-packages
(from nltk>=3.9->textblob) (4.67.1)
Downloading textblob-0.19.0-py3-none-any.whl (624 kB)
624.3/624.3 kB
1.7 MB/s eta 0:00:00ta 0:00:01
Installing collected packages: textblob
Successfully installed textblob-0.19.0

```

```

[18]: #data augmentation
X_train['status_enc'] = y_train # reattach target for augmentation

# Minor class augmentation
minor_classes = ['Personality disorder', 'Stress', 'Bipolar', 'Anxiety']
minor_subset = X_train[
    (X_train['status'].isin(minor_classes)) &
    ((X_train['stmt_len'] < 6) | (X_train['sentiment_score'].abs() > 0.5))
]

# Major class augmentation
major_classes = ['Normal', 'Depression', 'Suicidal']
major_subset = X_train[
    (X_train['status'].isin(major_classes)) &
    (X_train['sentiment_score'].abs() > 0.7)
]
major_sample = major_subset.sample(frac=0.1, random_state=42)

# Combine and backtranslate
df_to_augment = pd.concat([minor_subset, major_sample])
df_to_augment['statement'] = df_to_augment['statement'].apply(back_translate)

# Concatenate with original training data
X_train_aug = pd.concat([X_train, df_to_augment], ignore_index=True)

```

```
y_train_aug = pd.concat([y_train, df_to_augment['status_enc']],  
    ignore_index=True)
```

```
[19]: ros = RandomOverSampler(random_state=42)  
X_res, y_res = ros.fit_resample(X_train_aug[['statement']], y_train_aug)
```

```
[20]: X_train = X_res['statement'].values  
y_train = y_res  
X_test = X_test['statement'].values  
y_test = y_test.values
```

```
[21]: print(df.head())
```

```

                                statement    status
0                                oh gosh  Anxiety \
1  trouble sleeping confused mind restless heart ...  Anxiety
2  wrong back dear forward doubt stay restless re...  Anxiety
3          shifted focus something else still worried  Anxiety
4          restless restless month boy mean  Anxiety

   sentiment_score  stmt_len  status_enc
0             0.0000         2          0
1            -0.7269         7          0
2            -0.7351         9          0
3            -0.2960         6          0
4            -0.4939         5          0

```

```
[22]: # 8) TOKENIZATION + DATASET
```

```
tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')

class TextDataset(Dataset):
    def __init__(self, texts, labels, max_len=128):
        self.texts = texts
        self.labels = labels
        self.max_len = max_len

    def __len__(self):
        return len(self.labels)

    def __getitem__(self, idx):
        enc = tokenizer(
            self.texts[idx],
            padding='max_length',
            truncation=True,
            max_length=self.max_len,
            return_tensors='pt'
```



```

    )
    return {
        'input_ids': enc['input_ids'].squeeze(0),
        'attention_mask': enc['attention_mask'].squeeze(0)
    }, torch.tensor(self.labels[idx])

train_ds = TextDataset(X_train, y_train)
test_ds = TextDataset(X_test, y_test)
train_loader = DataLoader(train_ds, batch_size=16, shuffle=True)
test_loader = DataLoader(test_ds, batch_size=16)

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')

# 9) LOGISTIC REGRESSION ON BERT EMBEDDINGS

def extract_bert_features(dataloader):
    model = BertModel.from_pretrained('bert-base-uncased').to(device)
    model.eval()
    feats, labs = [], []
    with torch.no_grad():
        for (inp, lab) in tqdm(dataloader):
            inp = {k:v.to(device) for k,v in inp.items()}
            out = model(**inp).last_hidden_state[:,0,:].cpu().numpy()
            feats.append(out); labs.append(lab.numpy())
    return np.vstack(feats), np.concatenate(labs)

Xtr_feats, ytr_feats = extract_bert_features(train_loader)
Xte_feats, yte_feats = extract_bert_features(test_loader)

lr = LogisticRegression(max_iter=500)
lr.fit(Xtr_feats, ytr_feats)
y_pred = lr.predict(Xte_feats)
print("LogReg Accuracy:", accuracy_score(yte_feats, y_pred))
print(classification_report(yte_feats, y_pred))

cm = confusion_matrix(yte_feats, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.title("LogReg Confusion Matrix"); plt.xlabel("Pred"); plt.ylabel("True");
plt.show()

# 10) FINE-TUNED BERT CLASSIFIER

bert_clf = BertForSequenceClassification.from_pretrained(
    'bert-base-uncased',
    num_labels=df['status_enc'].nunique()
).to(device)

```

```

opt = AdamW(bert_clf.parameters(), lr=2e-5)
epochs = 3
sched = get_scheduler(
    'linear', optimizer=opt,
    num_warmup_steps=0,
    num_training_steps=len(train_loader)*epochs
)
loss_fn = nn.CrossEntropyLoss()

for epoch in range(epochs):
    bert_clf.train()
    total = 0
    for (inp, lab) in tqdm(train_loader):
        inp = {k:v.to(device) for k,v in inp.items()}
        lab = lab.to(device)
        out = bert_clf(**inp).logits
        loss = loss_fn(out, lab)
        opt.zero_grad(); loss.backward(); opt.step(); sched.step()
        total += loss.item()
    print(f"Epoch {epoch+1} Loss {total:.4f}")

bert_clf.eval()
preds, trues = [], []
with torch.no_grad():
    for (inp, lab) in tqdm(test_loader):
        inp = {k:v.to(device) for k,v in inp.items()}
        lab = lab.to(device)
        out = bert_clf(**inp).logits
        preds.extend(torch.argmax(out, axis=1).cpu().numpy())
        trues.extend(lab.cpu().numpy())

print("BERT Fine-Tune Report:")
print(classification_report(trues, preds))
cm = confusion_matrix(trues, preds)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.title("BERT Fine-Tune Confusion Matrix"); plt.xlabel("Pred"); plt.
    ylabel("True"); plt.show()

# 11) BERT + LSTM (Fine-tuned)

class BERT_LSTM(nn.Module):
    def __init__(self, hidden_dim=128, num_classes=7):
        super().__init__()
        self.bert = BertModel.from_pretrained('bert-base-uncased')
        self.lstm = nn.LSTM(
            input_size=768,
            hidden_size=hidden_dim,

```

```

        batch_first=True,
        bidirectional=True
    )
    self.fc = nn.Linear(hidden_dim*2, num_classes)

    def forward(self, input_ids, attention_mask):
        bert_out = self.bert(
            input_ids=input_ids,
            attention_mask=attention_mask
        ).last_hidden_state
        lstm_out, _ = self.lstm(bert_out)
        logits = self.fc(lstm_out[:,0,:])
        return logits

model3 = BERT_LSTM(hidden_dim=128, num_classes=df['status_enc'].nunique()).
    to(device)
opt3 = AdamW(model3.parameters(), lr=2e-5)

for epoch in range(epochs):
    model3.train()
    total = 0
    for (inp, lab) in tqdm(train_loader):
        inp = {k:v.to(device) for k,v in inp.items()}
        lab = lab.to(device)
        out = model3(**inp)
        loss = loss_fn(out, lab)
        opt3.zero_grad(); loss.backward(); opt3.step()
        total += loss.item()
    print(f"[LSTM] Epoch {epoch+1} Loss {total:.4f}")

model3.eval()
preds3, trues3 = [], []
with torch.no_grad():
    for (inp, lab) in tqdm(test_loader):
        inp = {k:v.to(device) for k,v in inp.items()}
        out = model3(**inp)
        preds3.extend(torch.argmax(out,1).cpu().numpy())
        trues3.extend(lab.numpy())

print("BERT+LSTM Report:")
print(classification_report(trues3, preds3))
cm = confusion_matrix(trues3, preds3)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.title("BERT + LSTM Confusion Matrix"); plt.xlabel("Pred"); plt.
    ylabel("True"); plt.show()

# 12) BERT + CNN (Fine-tuned)

```

```

class BERT_CNN(nn.Module):
    def __init__(self, num_classes, dropout=0.3):
        super(BERT_CNN, self).__init__()
        self.bert = BertModel.from_pretrained('bert-base-uncased')
        self.conv1 = nn.Conv1d(in_channels=768, out_channels=256,
        ↪kernel_size=3, padding=1)
        self.relu = nn.ReLU()
        self.pool = nn.AdaptiveMaxPool1d(1)
        self.dropout = nn.Dropout(dropout)
        self.fc = nn.Linear(256, num_classes)

    def forward(self, input_ids, attention_mask):
        bert_out = self.bert(input_ids=input_ids,
        ↪attention_mask=attention_mask).last_hidden_state
        x = bert_out.permute(0, 2, 1)
        x = self.conv1(x)
        x = self.relu(x)
        x = self.pool(x).squeeze(2)
        x = self.dropout(x)
        return self.fc(x)

model4 = BERT_CNN(num_classes=df['status_enc'].nunique()).to(device)
opt4 = AdamW(model4.parameters(), lr=2e-5)

for epoch in range(epochs):
    model4.train()
    total_loss = 0
    for (inp, lab) in tqdm(train_loader):
        inp = {k:v.to(device) for k,v in inp.items()}
        lab = lab.to(device)
        out = model4(inp['input_ids'], inp['attention_mask'])
        loss = loss_fn(out, lab)
        opt4.zero_grad(); loss.backward(); opt4.step()
        total_loss += loss.item()
    print(f"[CNN] Epoch {epoch + 1} Loss: {total_loss:.4f}")

model4.eval()
test_preds, test_labels = [], []
with torch.no_grad():
    for (inp, lab) in tqdm(test_loader):
        inp = {k:v.to(device) for k,v in inp.items()}
        out = model4(inp['input_ids'], inp['attention_mask'])
        test_preds.extend(torch.argmax(out, dim=1).cpu().numpy())
        test_labels.extend(lab.cpu().numpy())

print("BERT + CNN - Evaluation Report:")

```

```
print(classification_report(test_labels, test_preds))
cm = confusion_matrix(test_labels, test_preds)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.title("BERT + CNN Confusion Matrix"); plt.xlabel("Predicted"); plt.
    ylabel("True")
plt.show()
```

```
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```

```
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```

```
/opt/conda/lib/python3.10/site-packages/sklearn/linear_model/_logistic.py:465:
ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.
```

Increase the number of iterations (max_iter) or scale the data as shown in:

<https://scikit-learn.org/stable/modules/preprocessing.html>

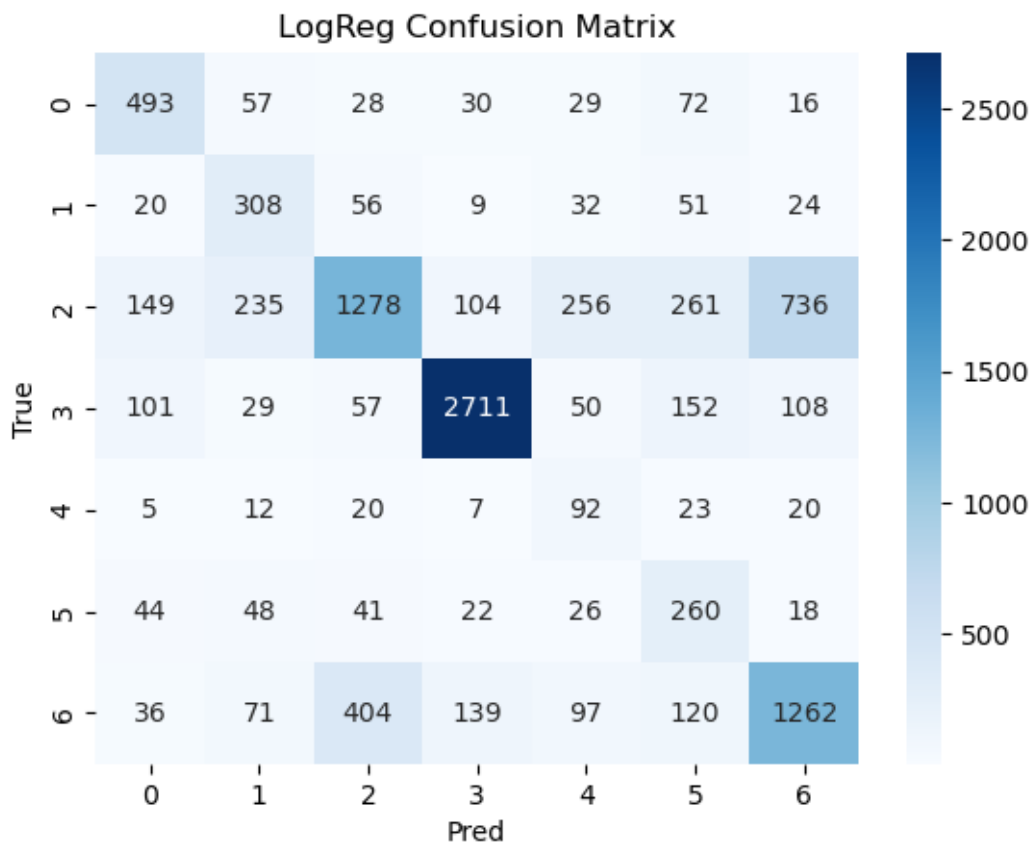
Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

```
n_iter_i = _check_optimize_result(
```

```
LogReg Accuracy: 0.6266757999804287
```

	precision	recall	f1-score	support
0	0.58	0.68	0.63	725
1	0.41	0.62	0.49	500
2	0.68	0.42	0.52	3019
3	0.90	0.85	0.87	3208
4	0.16	0.51	0.24	179
5	0.28	0.57	0.37	459
6	0.58	0.59	0.59	2129
accuracy			0.63	10219
macro avg	0.51	0.61	0.53	10219
weighted avg	0.68	0.63	0.64	10219



Some weights of BertForSequenceClassification were not initialized from the model checkpoint at bert-base-uncased and are newly initialized:

`['classifier.bias', 'classifier.weight']`

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

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Epoch 1 Loss 2820.7422

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Epoch 2 Loss 1133.3434

0%| | 0/5698 [00:00<?, ?it/s]

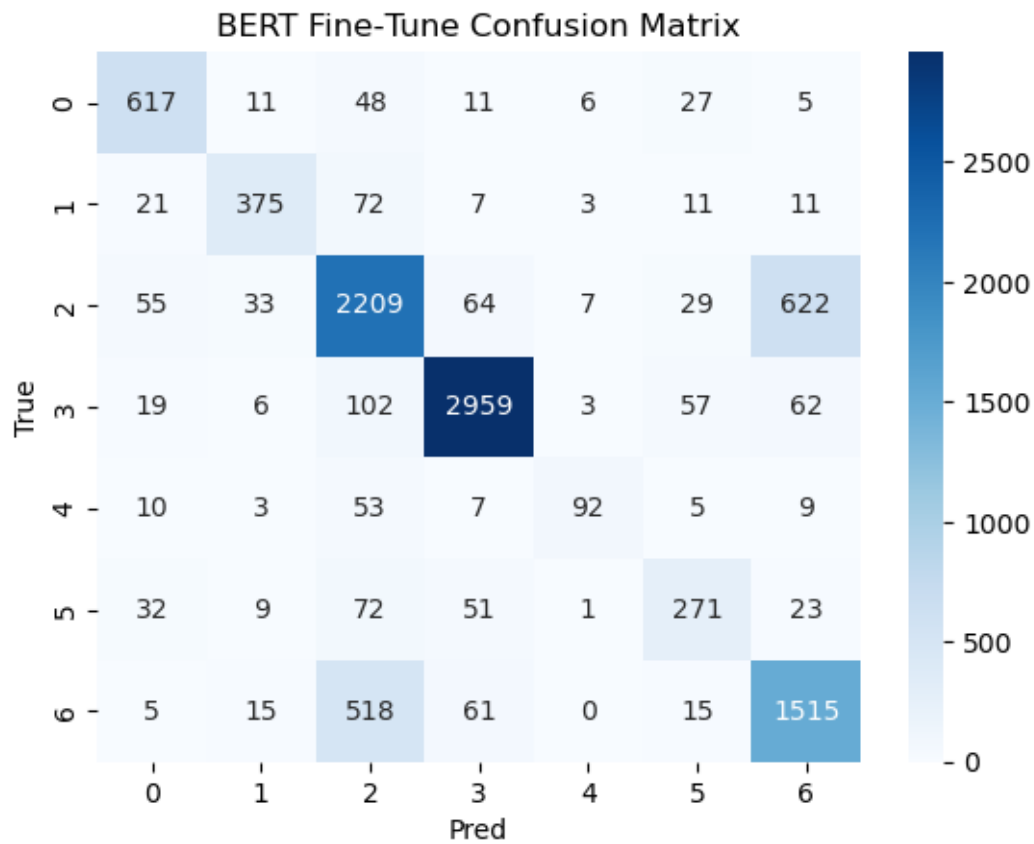
Epoch 3 Loss 689.6715

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BERT Fine-Tune Report:

	precision	recall	f1-score	support
0	0.81	0.85	0.83	725

1	0.83	0.75	0.79	500
2	0.72	0.73	0.73	3019
3	0.94	0.92	0.93	3208
4	0.82	0.51	0.63	179
5	0.65	0.59	0.62	459
6	0.67	0.71	0.69	2129
accuracy			0.79	10219
macro avg	0.78	0.72	0.75	10219
weighted avg	0.79	0.79	0.79	10219



```

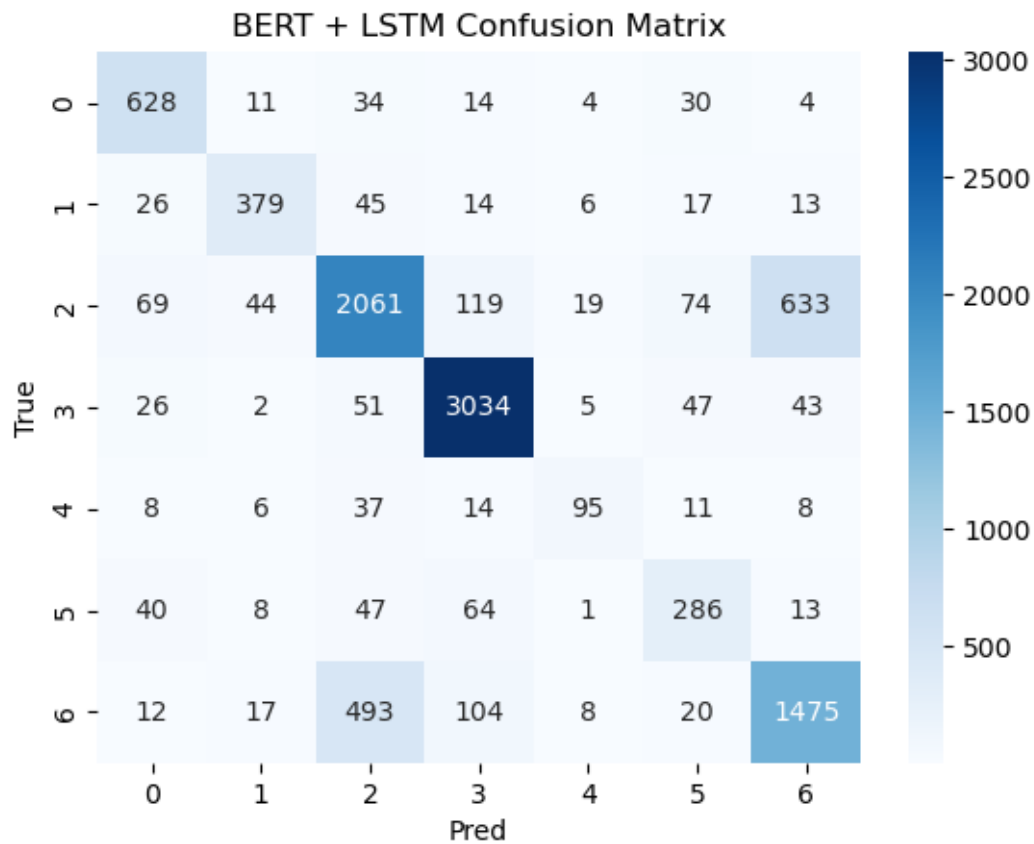
0%|          | 0/5698 [00:00<?, ?it/s]
[LSTM] Epoch 1 Loss 3006.0906
0%|          | 0/5698 [00:00<?, ?it/s]
[LSTM] Epoch 2 Loss 1311.2090
0%|          | 0/5698 [00:00<?, ?it/s]
[LSTM] Epoch 3 Loss 907.9289

```

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BERT+LSTM Report:

	precision	recall	f1-score	support
0	0.78	0.87	0.82	725
1	0.81	0.76	0.78	500
2	0.74	0.68	0.71	3019
3	0.90	0.95	0.92	3208
4	0.69	0.53	0.60	179
5	0.59	0.62	0.61	459
6	0.67	0.69	0.68	2129
accuracy			0.78	10219
macro avg	0.74	0.73	0.73	10219
weighted avg	0.78	0.78	0.78	10219



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[CNN] Epoch 1 Loss: 2940.6939

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[CNN] Epoch 2 Loss: 1288.6045

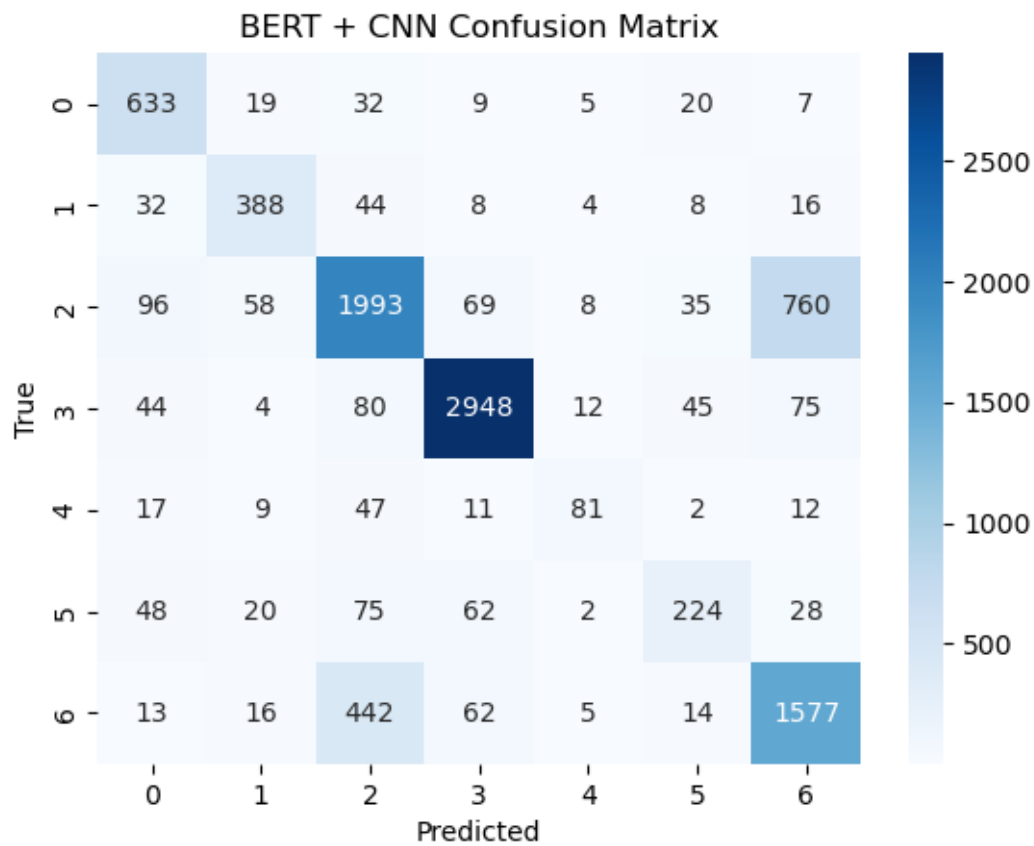
0%| | 0/5698 [00:00<?, ?it/s]

[CNN] Epoch 3 Loss: 878.3925

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BERT + CNN - Evaluation Report:

	precision	recall	f1-score	support
0	0.72	0.87	0.79	725
1	0.75	0.78	0.77	500
2	0.73	0.66	0.70	3019
3	0.93	0.92	0.92	3208
4	0.69	0.45	0.55	179
5	0.64	0.49	0.56	459
6	0.64	0.74	0.69	2129
accuracy			0.77	10219
macro avg	0.73	0.70	0.71	10219
weighted avg	0.77	0.77	0.77	10219



```
[31]: from torch.nn.functional import softmax
from sklearn.metrics import classification_report, confusion_matrix
import matplotlib.pyplot as plt
import seaborn as sns

# Set models to evaluation mode
bert_clf.eval()
model3.eval()
model4.eval()

ensemble_preds = []
true_labels = []

with torch.no_grad():
    for inp, lab in tqdm(test_loader):
        inp = {k: v.to(device) for k, v in inp.items()}
        lab = lab.to(device)

        # Get logits from each model
        logits_bert = bert_clf(**inp).logits
        logits_lstm = model3(**inp)
        logits_cnn = model4(inp['input_ids'], inp['attention_mask'])

        # Convert logits to probabilities
        probs_bert = softmax(logits_bert, dim=1)
        probs_lstm = softmax(logits_lstm, dim=1)
        probs_cnn = softmax(logits_cnn, dim=1)

        # Soft voting: average the probabilities
        avg_probs = (0.5 * probs_bert + 0.5 * probs_lstm)

        # Final predictions
        preds = torch.argmax(avg_probs, dim=1)

        ensemble_preds.extend(preds.cpu().numpy())
        true_labels.extend(lab.cpu().numpy())
```

```
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```

```
[32]: # Show classification report
print("Soft Voting Ensemble Performance:")
print(classification_report(true_labels, ensemble_preds, target_names=le.
↪classes_))
```

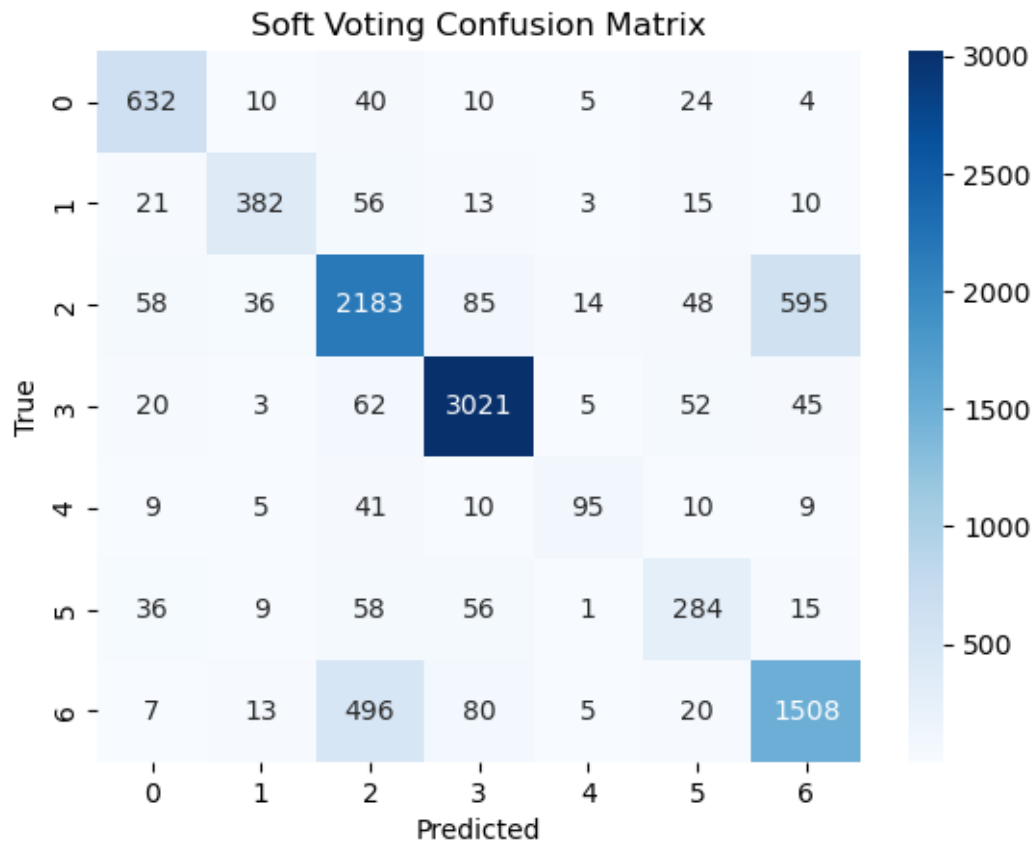
```

# Confusion matrix
cm = confusion_matrix(true_labels, ensemble_preds)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.title("Soft Voting Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("True")
plt.show()

```

Soft Voting Ensemble Performance:

	precision	recall	f1-score	support
Anxiety	0.81	0.87	0.84	725
Bipolar	0.83	0.76	0.80	500
Depression	0.74	0.72	0.73	3019
Normal	0.92	0.94	0.93	3208
Personality disorder	0.74	0.53	0.62	179
Stress	0.63	0.62	0.62	459
Suicidal	0.69	0.71	0.70	2129
accuracy			0.79	10219
macro avg	0.77	0.74	0.75	10219
weighted avg	0.79	0.79	0.79	10219



[]: